

Module 3: Economic Impacts of Monetary Policy

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Lesson 3-0 Module 3 Information

[Lesson 3-0.1: Overview](#)

Module overview

- Fed's dual mandate
 - Maximum employment
 - Stable prices
- Monetary policy actions
 - From interest rates...
 - ...to broader economy

Hello and welcome, I hope you're doing well from wherever it is you're joining us. The purpose of the Federal Reserve is to achieve maximum employment and stable prices. This is called the Fed's dual mandate. In an ideal world, the economy would have minimal inflation, and everybody would have a job, however, this is not always the case. Okay, but how do the Fed monetary policy actions affect the economy? This is the goal of this module; we will look at connections between monetary policy actions that bring about changes in the most important macroeconomic outcomes. We will also discuss how these changes happen; we will start with the idea of maximum employment. While the dual mandate is stated in terms of jobs, many central banks, including the Federal Reserve, talk a lot about economic growth. To explain why we will first study the relationship between gross domestic product or GDP and unemployment.

Fed and the economy

- Easy monetary policy can
 - Grow GDP and employment
 -but increase inflation
- How to design monetary policy?
 - Taylor Rule

We will then see how decreases in the Fed's target interest rate can increase GDP and therefore decrease unemployment. As part of this discussion, we will see what the goal of maximum employment means in the context of the United States. Next, we will study the link between unemployment, wages, and inflation, we will see that there is a tradeoff between unemployment and inflation. By easing monetary policy, a central bank can lower unemployment at least in the short run, but this may come at the cost of higher inflation. This tradeoff between unemployment and inflation is central to decision making at the Federal Reserve. We will see how this trade-off is embedded in the monetary policy rules such as the celebrated Taylor Rule. These rules dictate exactly how interest rates should be set depending on the state of the economy. With this tradeoff between inflation and unemployment in mind, we will then study how monetary policy works. That is through which channels to changes in the interest rate affect economic growth and thereby inflation and unemployment. We start with a high-level overview that shows you the effects of monetary policy on the economy as a whole. We then drill down on how monetary policy affects credit supply by banks and therefore money supply in the economy. The next lesson discusses how monetary policy affects firms and households which in turn affects investment and consumption decisions. Both consumption and investment are part of the GDP, and these decisions are going to affect unemployment. Monetary policy can mitigate harmful boom and bust in the economic cycle by raising the interest rates during expansions and reducing the interest rate during contractions. This is what Central banks hope to achieve.

Limits of monetary policy

- Supply-side shortages
- Industries in decline

The last two lessons are case studies that illustrates the limits of monetary policy. We will see that monetary policy can do little to offset supply-side shocks such as spikes in oil prices. It cannot do much to support employment in industries that are in permanent decline due to structural changes in the economic environment. In some, the previous module discussed which monetary policy tools are available to central banks and how they are affecting the interest rates. In this module, we look at how interest rate changes affect the economy, and how this helps the Federal Reserve to choose its dual mandate of stable prices and maximum employment.

Lesson 3-1: The Link Between Inflation and Unemployment

Lesson 3-1.1. Understanding Economic Growth and Unemployment



Learning Objectives

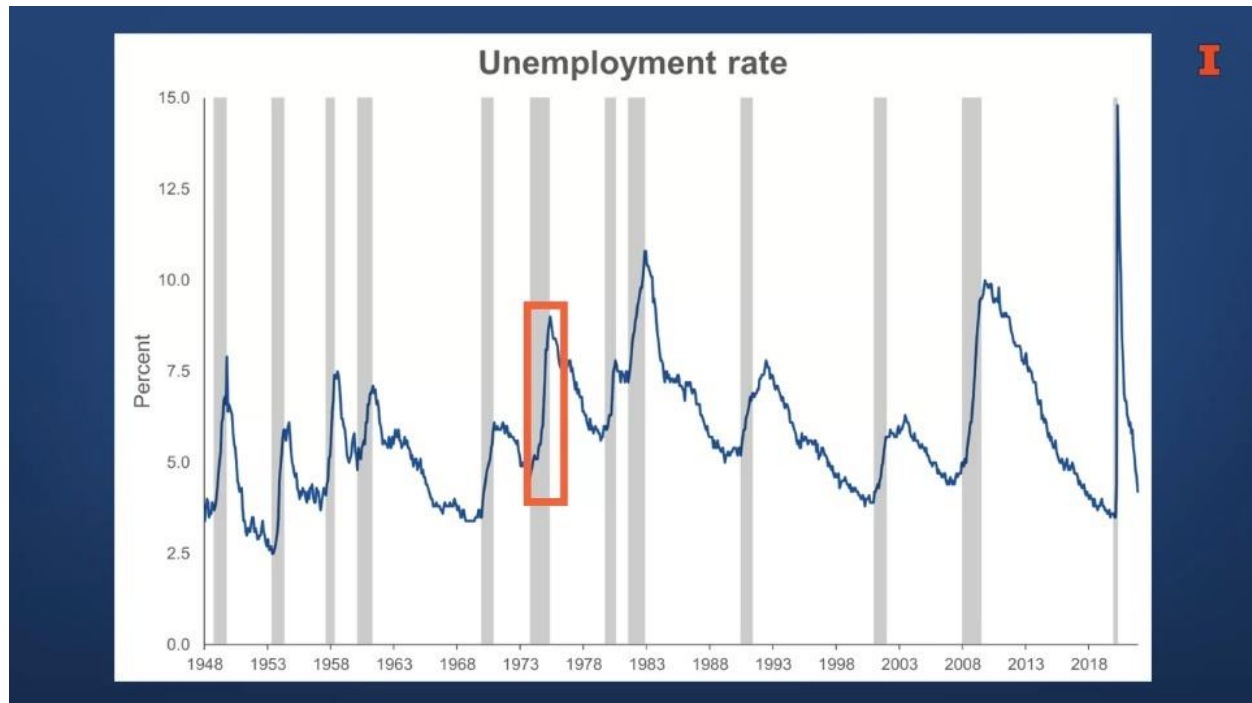
Understand “unemployment” terminology:

1. “Natural rate”
2. “Full employment”
3. “Maximum employment”

Define gross domestic product (GDP)

Relation between GDP and unemployment

Hello and welcome to this lecture on economic growth and unemployment. Remember, the Federal Reserve uses monetary policy to achieve stable prices and maximum employment. But what exactly do we mean by maximum employment? In this class, we will examine unemployment. We will introduce some new definitions of unemployment. The natural rate of unemployment, full employment, maximum employment. We will understand what this terminology means and why it is needed. We will also define gross domestic product or GDP. This is a measure of total output for an economy. We will see the relationship between the unemployment rate and GDP growth in the data. Finally, we will understand why the Federal Reserve does not try to reduce the unemployment rate to zero.



Let's start with the unemployment data. An individual is unemployed if they do not have a job and they are seeking employment. The unemployment rate is the share of the labor force that is currently unemployed. In the graph you can see the US unemployment rate in the modern era. The shaded areas in the graph indicate recessions. You can see that unemployment increases rapidly during a recession. For instance, from October 1973 to May 1975, the unemployment rate nearly doubled. It increased from 4.6% to 9%, at the same time, unemployment falls rapidly after the end of a recession. By March 1976, the unemployment rate had already dropped to 7.6%.

Gross Domestic Product

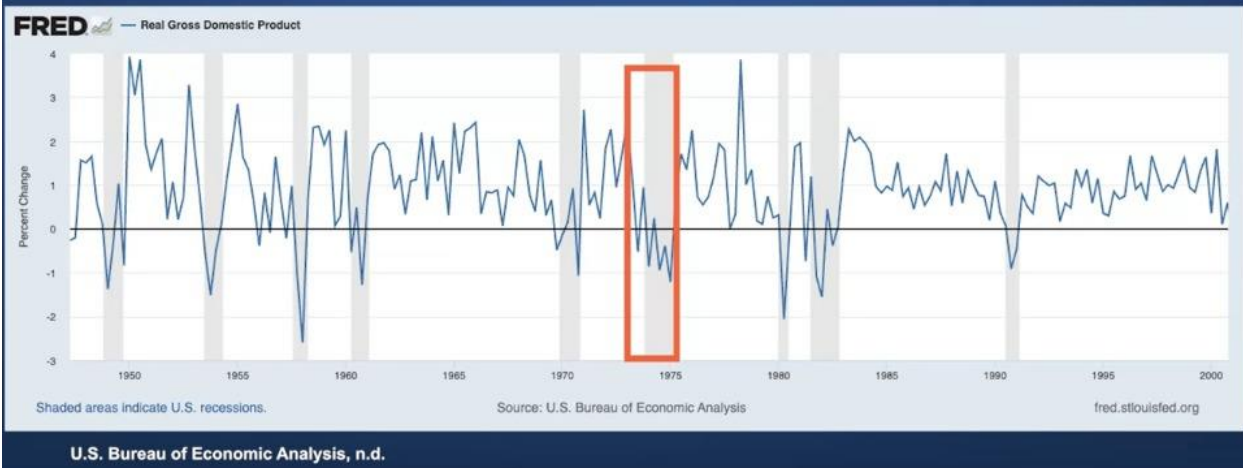
GDP: dollar value of national output

Real GDP: inflation-adjusted GDP



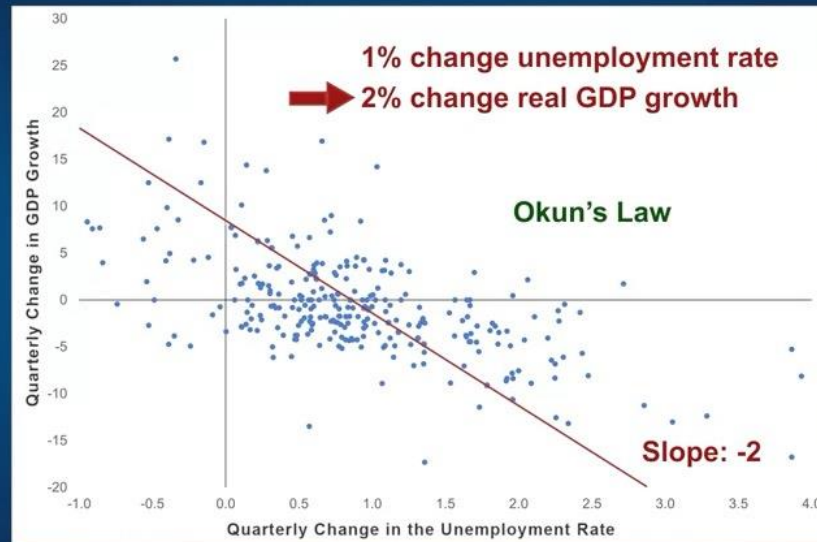
Now let's turn to our measure of economic growth. Gross Domestic Product. GDP is the total dollar value of all goods and services produced by an economy. Real GDP adjusts this measure for inflation. Real GDP in the US has been growing over time. However, during recession it falls, this means the economy contracts. Let's look at this in the data.

Quarterly Growth in Real Gross Domestic Product (GDP)



The graph shows quarterly percent changes of growth in real GDP. During recessions, the shaded areas real GDP growth is negative. You can see that GDP fell by as much

as 1% during each quarter of 1974. What is the relationship between unemployment and real GDP growth? During recession unemployment goes up and GDP goes down, during expansion, the opposite happens, this suggests a negative correlation.



Let's combine the previous two charts to see how this works. Each dot represents a quarter. On the x axis, we have the growth in unemployment relative to the prior quarter. And on the y axis we have the GDP growth for that same quarter. We can see the negative correlation in the data, this correlation is indicated by the downward sloping red line. Based on historical US data, the slope of the line is about -2. This empirical fact was first stated by the economist Arthur Okun, it is known as Okun's Law. Why do monetary policymakers find Okun's Law useful? Because Okun's Law gives some guidance about how much unemployment will decline when they cut the interest rate to stimulate the economy. Once you have an estimate for how much real GDP is going to increase after an interest rate cut, then you also know by how much unemployment is going to decline. Understanding the relationship between these macroeconomic variables enables the Fed to better target monetary policy. Let's now ask a different question.

Natural Rate of Unemployment

Unemployment rate never = 0

Why? Job search

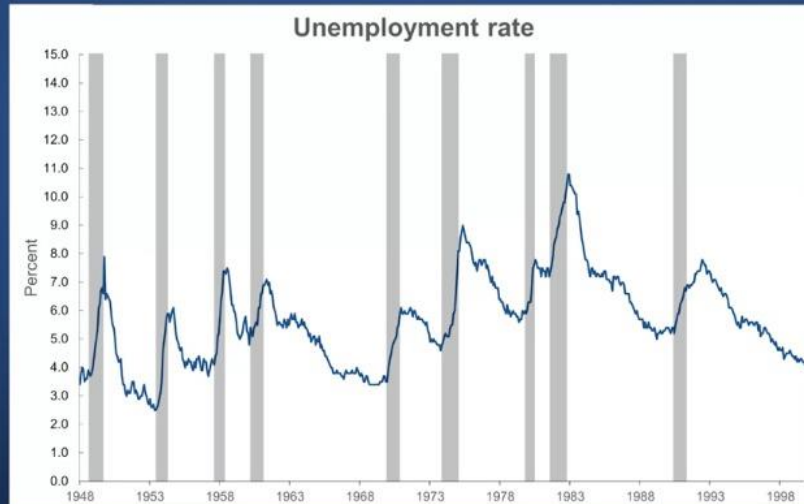
New technology

Unemployment benefits

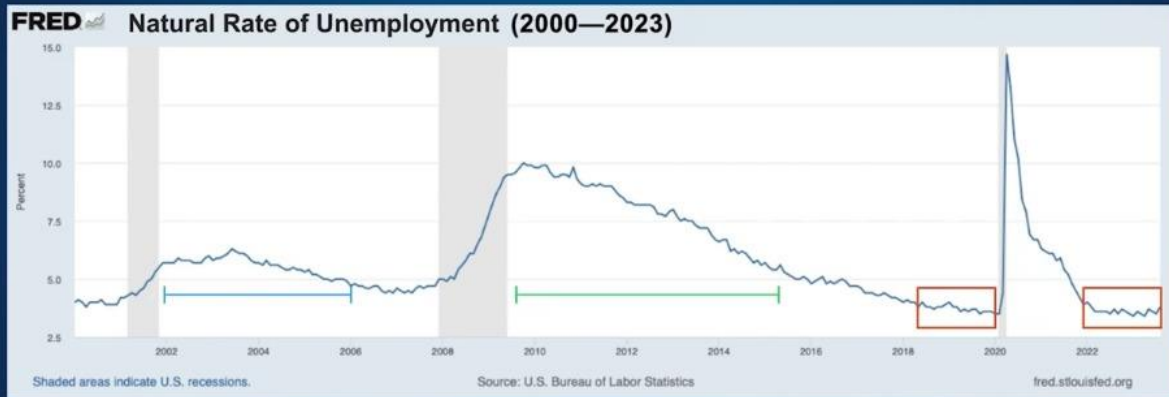


You might be thinking, why can't central banks reduce the unemployment rate to zero? The answer is that there is a natural rate of unemployment. This means that even if the economy is at peak performance, some workers will still be unemployed. One reason for this natural rate of unemployment is that it takes time to find a suitable job. A job that is a good fit with your schedules or skill sets. This is true for individuals that recently left education, recently lost a job, or those relocating. Other factors creating positive unemployment include new technologies that make all skills obsolete and force reallocation. Just consider what autonomous driving will mean for truck drivers. Higher unemployment benefits may also give unemployed more time to find a better fitting job. Let's look again at the time series of the unemployment rate in the US.

Estimating the Natural Rate of Unemployment



Roughly speaking, we can estimate the natural rate based on unemployment during good economic times. Estimates suggest that in the 1950s and 1960s, the natural rate of unemployment was 4%. For the period between 1970 and 1983, the natural rate was estimated to be 6%, and from 1984 to 2000, the estimate went down to 5.5%. The chart also shows large cyclical fluctuation in unemployment. Notably, during recessions, unemployment exceeds the natural rate. This is where monetary policy can be effective.



U.S. Bureau of Economic Analysis, n.d.

One last question what is the natural rate of unemployment today? Let's look at unemployment in the new millennium. After the 2001 recession, the unemployment rate recovered only slowly as compared with the recessions in the second half of the 20th century. Similarly, after the 2008 financial crisis, the unemployment rate did not recover as quickly as expected. This prompted speculation that the natural rate of unemployment had increased. However, by December 2019, close to the end of the longest economic expansions in the US the unemployment rate was 3.6%. Likewise, after the COVID pandemic, the unemployment rate was historically low. This suggests that the natural rate of unemployment may have decreased. Overall, it is hard to say, new technologies or changes in the labor market might play a role. However, we need more time before we can decide.

Summary

Okun's Law: Changes in unemployment and GDP are negatively correlated.

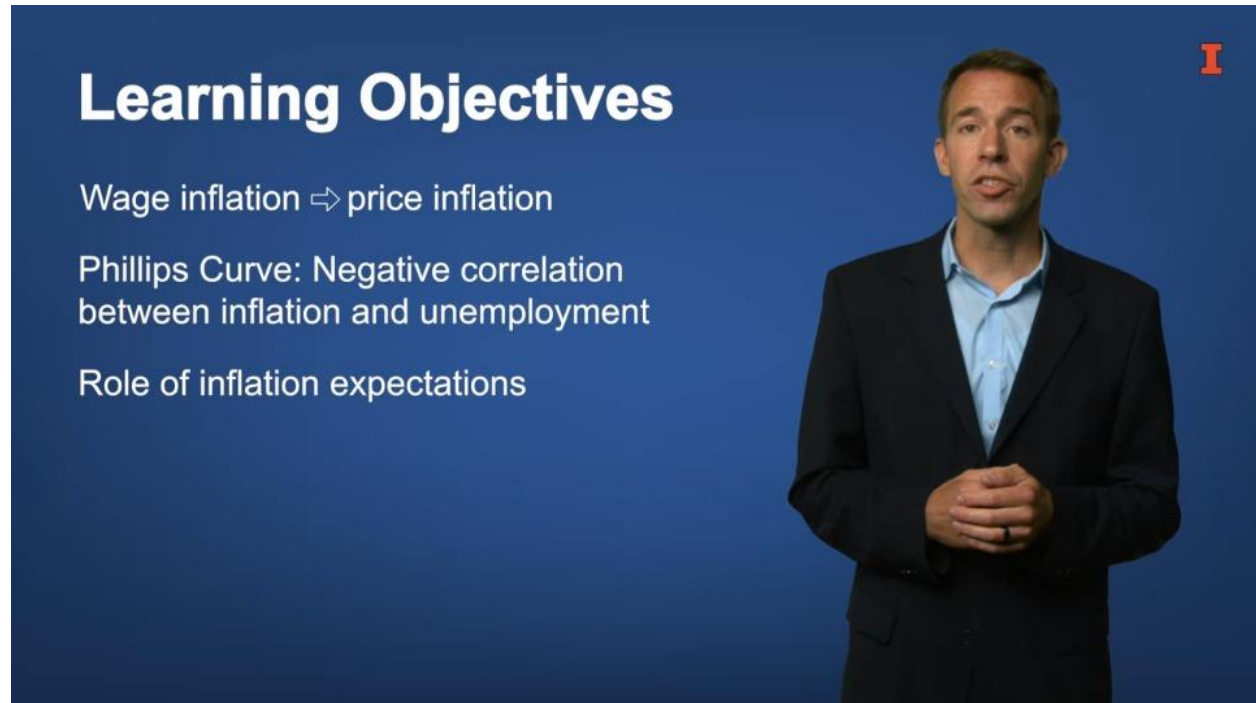
Natural rate of unemployment $\gg 0$



Let's wrap up, we covered two concepts in this session. First, we saw that changes in unemployment and real GDP are negatively correlated in the data, this is called Okun's law. This relationship guides monetary policy decisions. Second, even when the economy is growing at full capacity, unemployment will be greater than zero, this is called the natural rate of unemployment.

Lesson 3-1: The Link Between Inflation and Unemployment

[Lesson 3-1.2. Connecting Unemployment and Inflation](#)



Learning Objectives

- Wage inflation $\Rightarrow$ price inflation
- Phillips Curve: Negative correlation between inflation and unemployment
- Role of inflation expectations

Hello, and welcome to this lecture on the relationship between wages, unemployment and inflation. Inflation targeting is central to monetary policy. In this class, we will discuss how labor costs or wage inflation, feeds into consumer price inflation. We will see how wages and inflation move together. We will then examine the relationship between inflation and unemployment. For that, we will introduce the idea of the Phillips Curve. This is one of the most famous charts in economics. You will see, how it describes the tradeoff between inflation and unemployment. Less inflation usually comes at a cost of higher unemployment. Finally, we will understand how changes in inflation expectations can shift the Phillips Curve.

Inflation, Defined

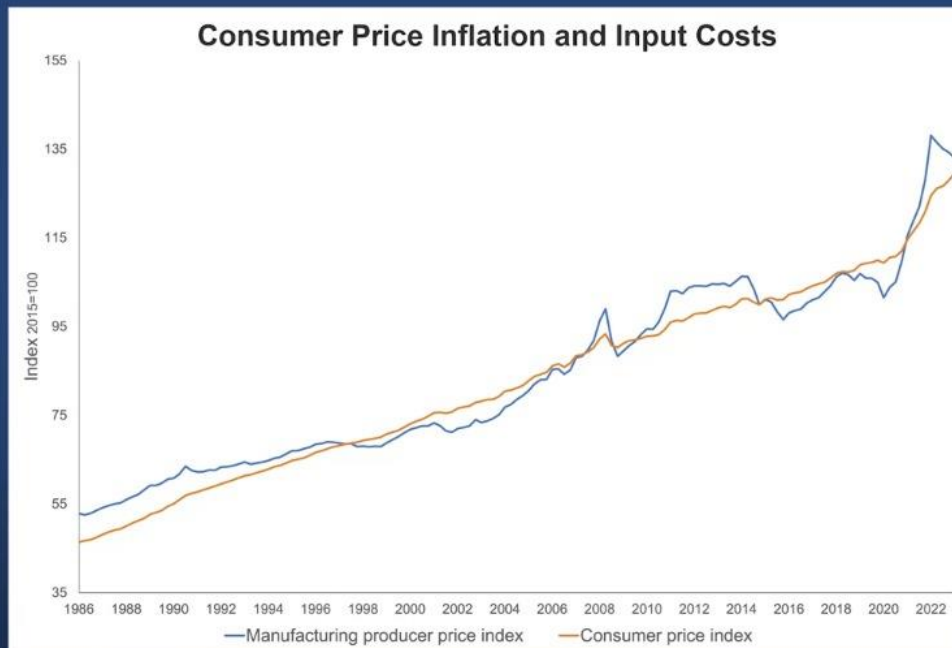
Consumer price inflation (CPI)
Growth in price of consumption bundle

Producer price inflation (PPI)
Growth in input prices

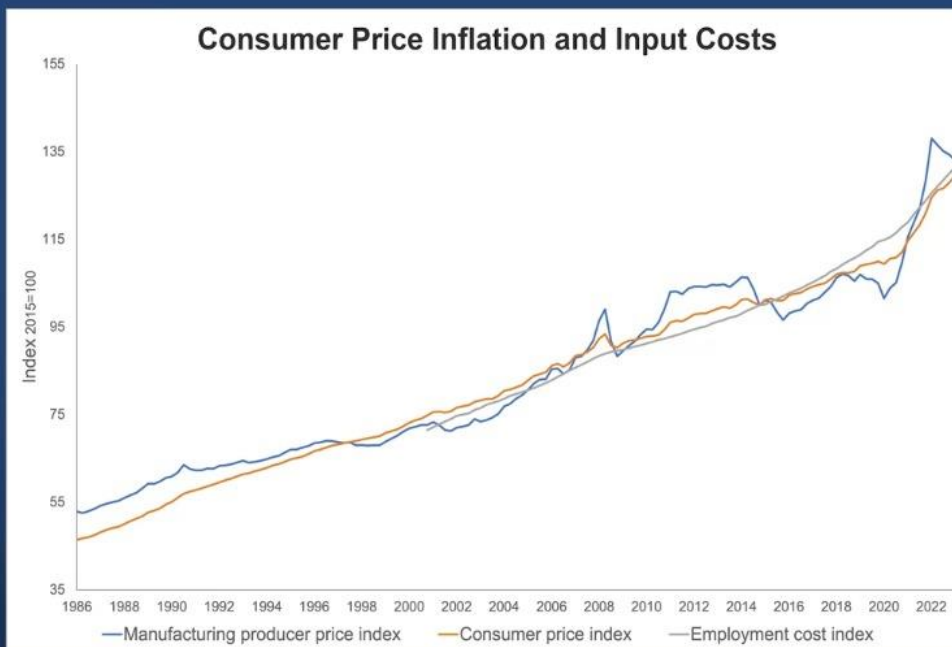
Higher PPI $\Rightarrow$ higher CPI



Economists, policymakers, or the press, talk about inflation all the time. They are usually referring to consumer price inflation, or CPI. This refers to growth in the price of a bundle of consumer goods. You may wonder, what causes consumer prices to go up. One answer is input prices. We all realize that fueling a car is more expensive when crude oil prices are high. More generally, when input prices rise, the cost of producing goods and providing services, goes up too. Put differently, higher producer price inflation, can lead to an increase in the CPI. Let's look at some data on this.

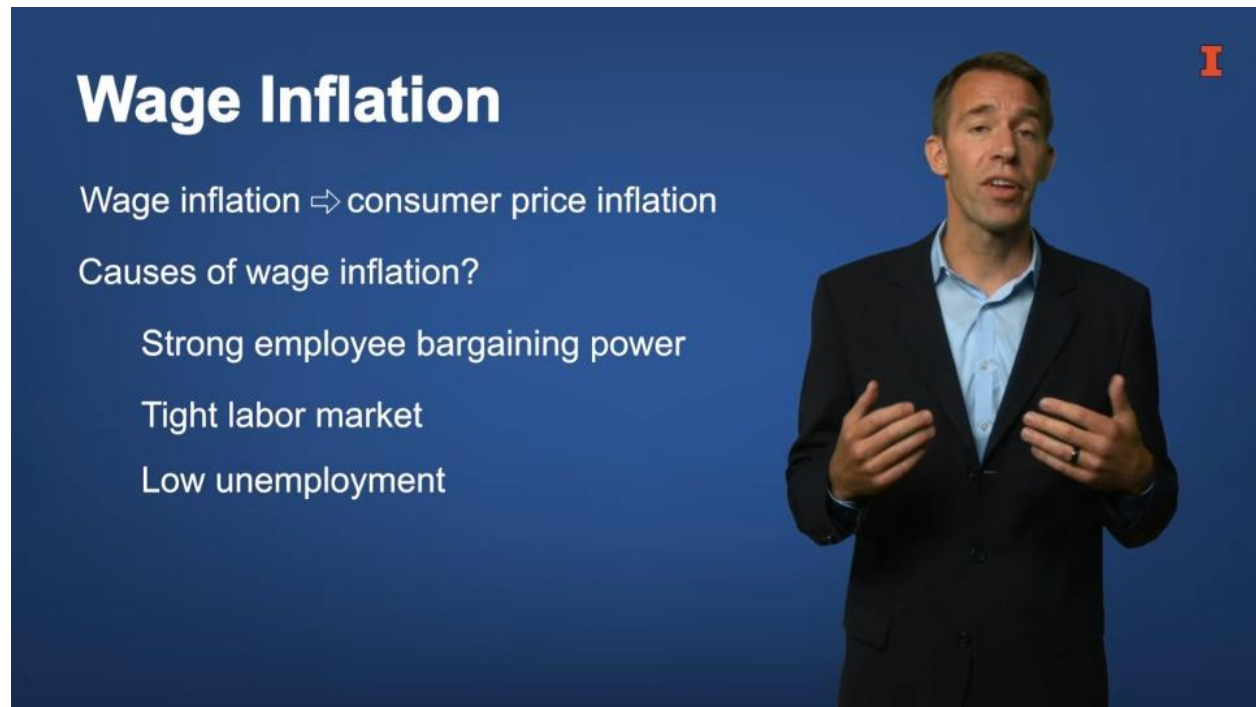


This graph shows consumer price inflation, and producer price inflation since the 1980s. We can see that, while the producer price index is more volatile, the two series track each other. Higher producer prices, lead to higher consumer goods prices.



What is the biggest cost component of most industries? Labor expenses let's add an employment cost index to this graph. This employment cost index includes both wages and benefits. The series only begins in 2000, so we must restrict the time frame. We

see again that employment costs and consumer price inflation, track each other. This suggests that increases in wages lead to consumer price inflation.



Wage Inflation

Wage inflation $\Rightarrow$ consumer price inflation

Causes of wage inflation?

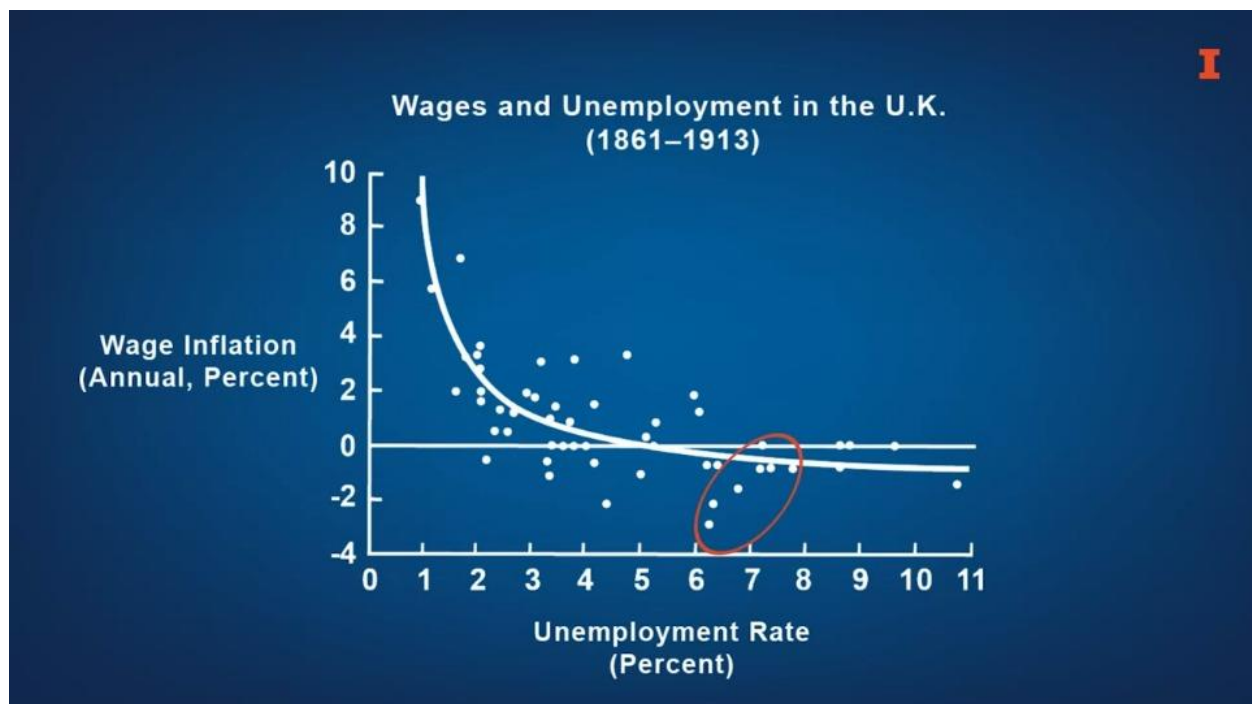
- Strong employee bargaining power
- Tight labor market
- Low unemployment

The slide features a man in a dark suit and light blue shirt speaking, with his hands gesturing. A small red 'I' logo is in the top right corner of the slide area.

Let's push this reasoning one step further. When do wages increase most? What causes wage inflation? Wages go up usually when employees have the most bargaining power. This is when unemployment is low, and there's less competition for jobs. When unemployment is high, employers do not have to pay higher wages to attract job applicants.

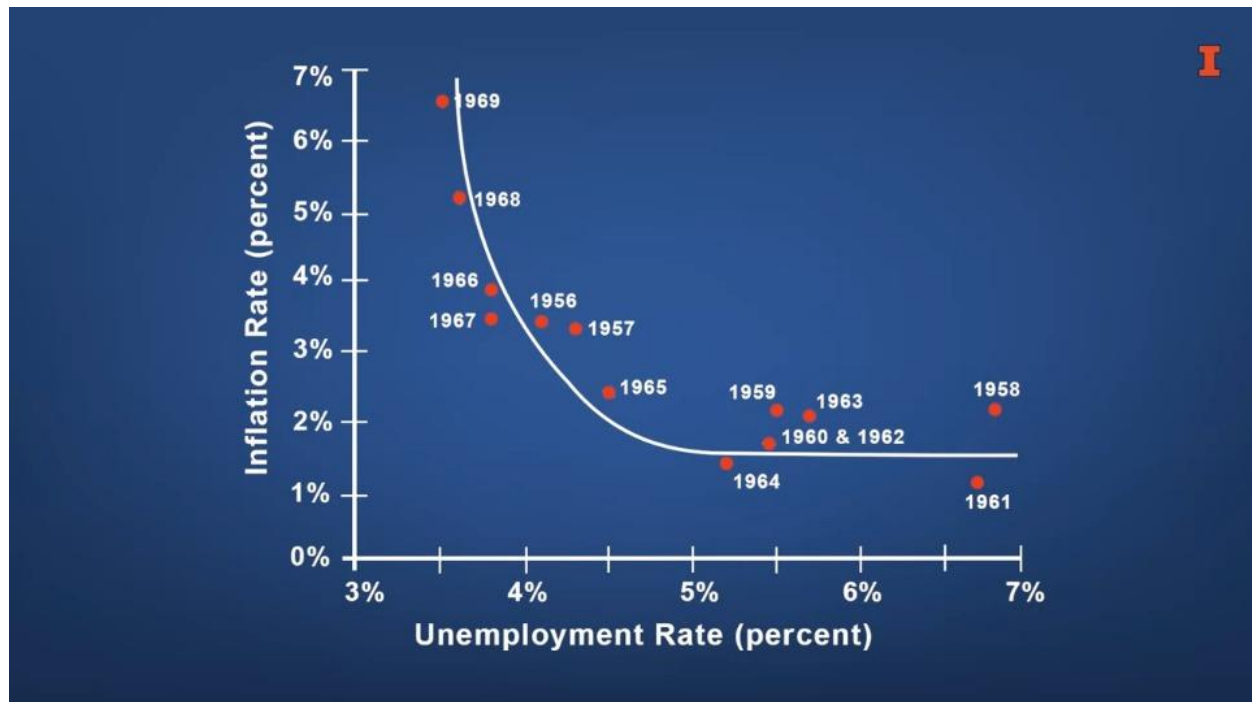


This relationship was first noted by Arthur Phillips. He plotted unemployment against wage growth in the United Kingdom. He found a negative correlation between the two outcomes.



When unemployment is low, wages tend to go up. When unemployment is high, wages tend to shrink. This relationship is often called the Phillips Curve. Nowadays, the term Phillips curve, is used to describe the relationship between unemployment and

consumer price inflation. We saw earlier, how CPI and wage growth track each other, so this is understandable. Let's examine the Phillips Curve in the United States, in the more recent years.



Here's the data from the 1960s. As before, we have the unemployment rate on the X axis, but now we have consumer price inflation, on the Y axis. We can see the negative relationship between inflation and unemployment holds.

Implications of the Phillips Curve

Trade-off between inflation and unemployment

Any policy that reduces unemployment may increase inflation

Conflict in Fed's dual mandate



Why should we care? What are the implications for policymakers? This graph suggests a clear tradeoff between unemployment and inflation. This means, that any government policy that reduces unemployment may increase inflation. This matters for monetary policy. Remember, stable prices and maximum employment are the Federal Reserve's mandate. These goals are in conflict. Achieving both goals at the same time is a balancing act. In practice, we can see key decision makers at the Fed, thinking about the Phillips curve.

June 11, 2008

Lessons for Central Bankers from a Phillips Curve Framework

Vice Chairman Donald L. Kohn

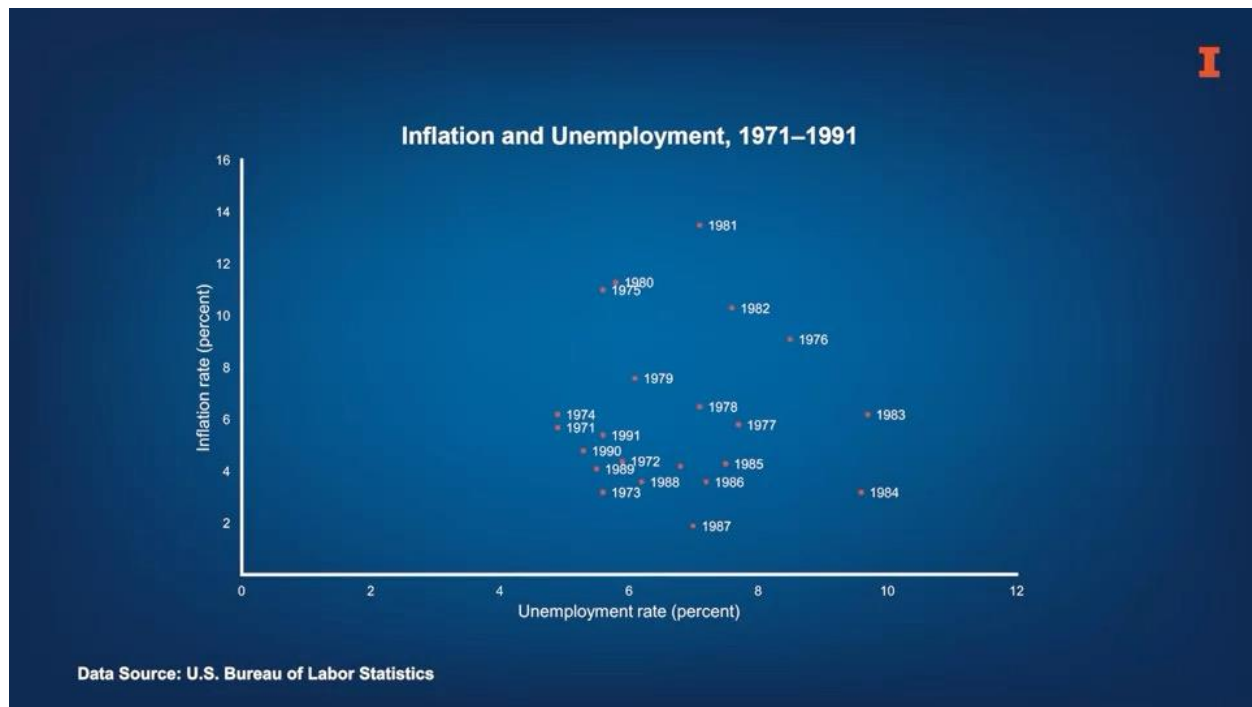
At the Federal Reserve Bank of Boston's 53rd Annual Economic Conference, Chatham, Massachusetts

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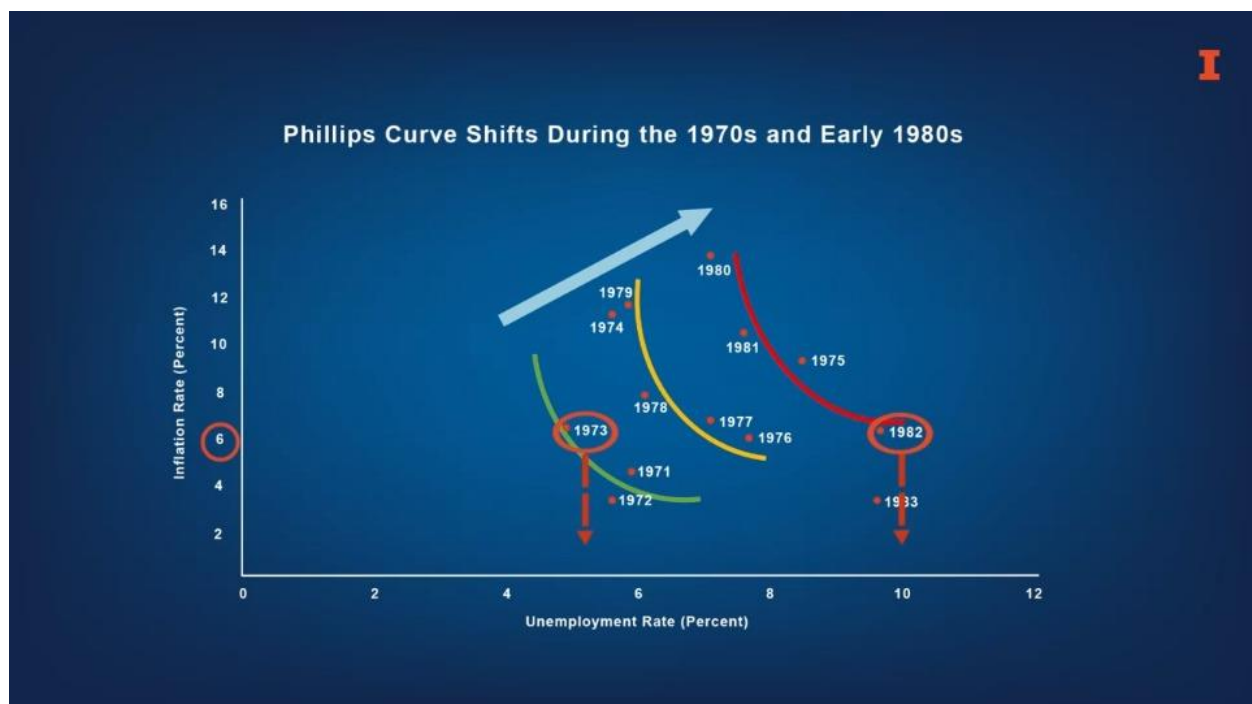
An economic model of inflation is an indispensable input to monetary policy deliberations. A model in the Phillips curve tradition remains at the core of how most academic researchers and policymakers—including this one—think about fluctuations in inflation; indeed, alternative frameworks seem to lack solid economic foundations and empirical support. But the modern Phillips curve differs substantially from versions in use several decades ago; policymakers and academics alike are now attuned to the importance of expectations, the possibility of structural change, and the uncertainty that surrounds our understanding of the dynamics of wage and price adjustment. Moreover, the link between inflation and resource utilization often emphasized in a Phillips curve framework accounts for only a modest part of inflation fluctuations. My comments today will focus on how the lessons from recent research on the Phillips curve are helping me think about the influence of fluctuations in the prices of commodities, such as oil, on the outlook for inflation and the appropriate policy responses to such developments.¹

Kohn, 2008

For instance, in 2008, the then Vice Chairman of the Board of Governors of the Federal Reserve System, Donald Cohn, stated that, an economic model of inflation is an indispensable input to monetary policy deliberations. A model in the Phillips curve tradition remains at the core of how most academic researchers and policymakers, including this one, think about fluctuation in inflation. Indeed, alternative frameworks seem to lack solid economic foundations, and empirical support. Are we done? Well, not quite.

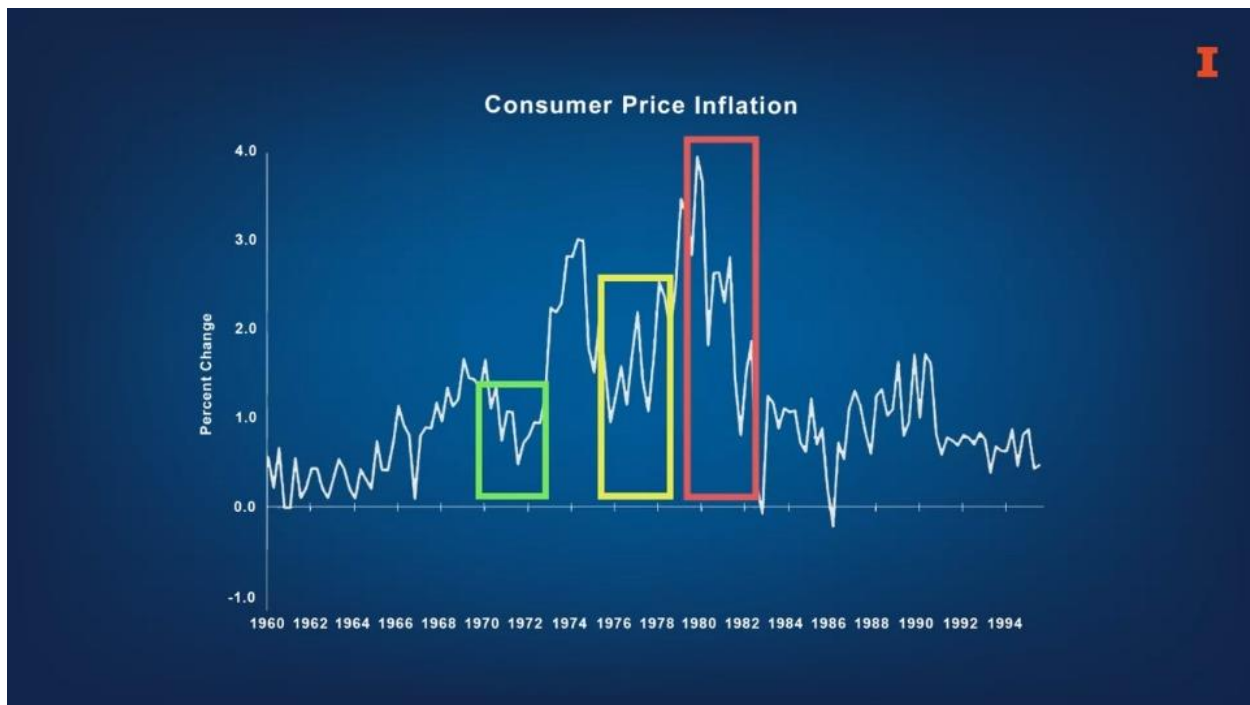


In the 1960s and early 1970s, it was thought that the relationship between unemployment and inflation was stable. However, over the next two decades, a different pattern emerged. Here's the data for the 1970s and 1980s. Unfortunately, the negative correlation seems to have vanished.

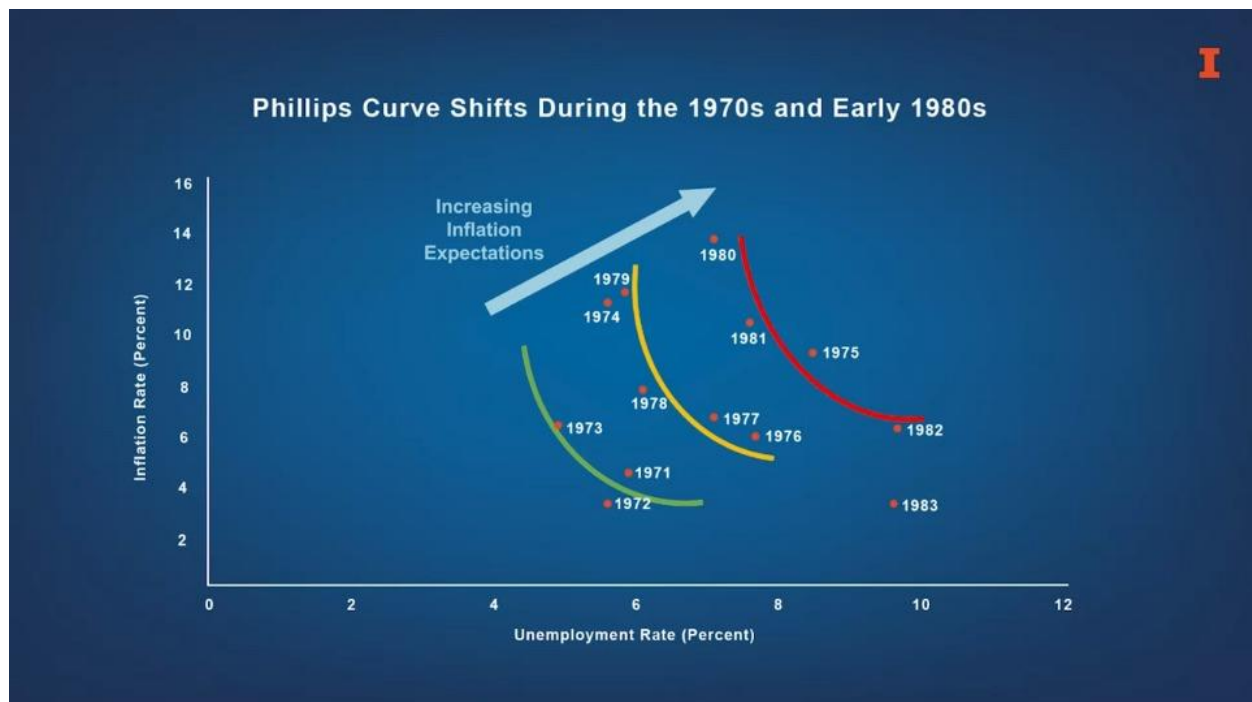


However, when you squint a little, the usual pattern seems to emerge. It looks like, that there is not one, but three relationships. One for the early 1970s, one for the late 1970s,

and one for the early 1980s. It looks like there are three Phillips curves. Weird, right? These curves also seem to be shifting outwards over time. An outward shift is not a good thing. The further the Phillips curve shifts out, the more unemployment is needed to keep inflation in check. For example, let's say, we want to hold inflation at 6%. In 1973, this would require an unemployment rate of 5%. However, by 1982, unemployment would have reached 10% to achieve the same inflation rate. While the 1970s and 1980s cast out on the overall relationship between inflation and unemployment, the data indicates that, at least in the short run, the Phillips curve works. We now examined, what determines these short run Phillips curves, in the 1970s and 1980s.



Let's look at consumer price inflation in the United States, between 1960 and 1995. Clearly, during the early 1970s, late 1970s, and early 1980s, consumer price inflation was higher and increasing as compared with the 1960s. Hence, a plausible explanation for the outward shift of the Phillips curve, could be an upward shift in inflation expectations, from the early 1970s to the early 1980s. Why do expectations about future inflation matter? This is not hard to see. Suppose your employer announces a 5% wage increase. This sounds like a great increase. However, what if consumer price inflation is currently running at 10%, and is expected to remain at 10% next year? This means, that your income adjusted for inflation, your real wage falls by 5%. That's not great news. Only real wage increases raise living standards. This suggests that when workers expect higher inflation they demand even higher wages. This alters the relationship between wages and unemployment. And that has knock on effects for the relationship between unemployment and consumer price inflation.



The increase in inflation expectation explains the outward shift of the short run Phillips curves, that we saw in the 1970s and 1980s. It also explains why policymakers keep a close eye on inflation expectations. They try to keep inflation expectations well anchored.

Summary

Wages drive inflation

Phillips Curve: trade-off between unemployment and inflation

Inflation expectations shift Phillips Curve

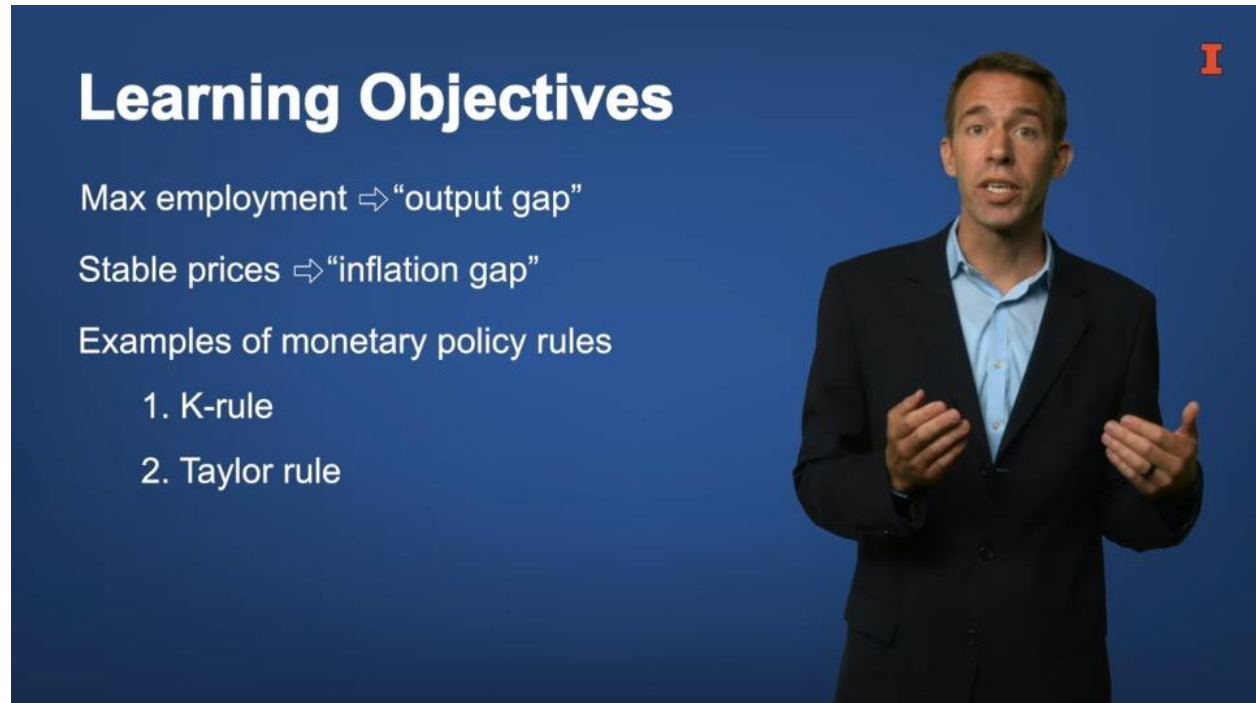


Let's wrap up. We have covered three big ideas in this session. First, wages and

consumer price inflation, move together. Second, there is a tradeoff between inflation and unemployment, modeled in the Phillips curve. Third, changes in inflation expectation, can shift the Phillips curve.

Lesson 3-1: The Link Between Inflation and Unemployment

Lesson 3-1.3. Monetary Policy Rules



Learning Objectives

- Max employment $\Rightarrow$ “output gap”
- Stable prices $\Rightarrow$ “inflation gap”
- Examples of monetary policy rules
 1. K-rule
 2. Taylor rule

Hello, and welcome to this class on the design of monetary policy rules. Nowadays, central banks set the interest rate to achieve their goals. In the Federal Reserve's case, they adjust the federal funds rate, but the Fed faces a tradeoff between their goals of stable prices and maximum employment. Remember the Phillips Curve? In this session, we will discuss how this tradeoff is approached in practice. We will introduce the idea of an output gap and an inflation gap. We will then ask what central banks should do about these gaps. We will examine Milton Friedman's K-rule, which says that monetary policy should not respond to economic conditions. Then we will examine the famous Taylor Rule that ties interest rates to gaps or fluctuations in inflation and employment.

Goal #1: Maximum Employment

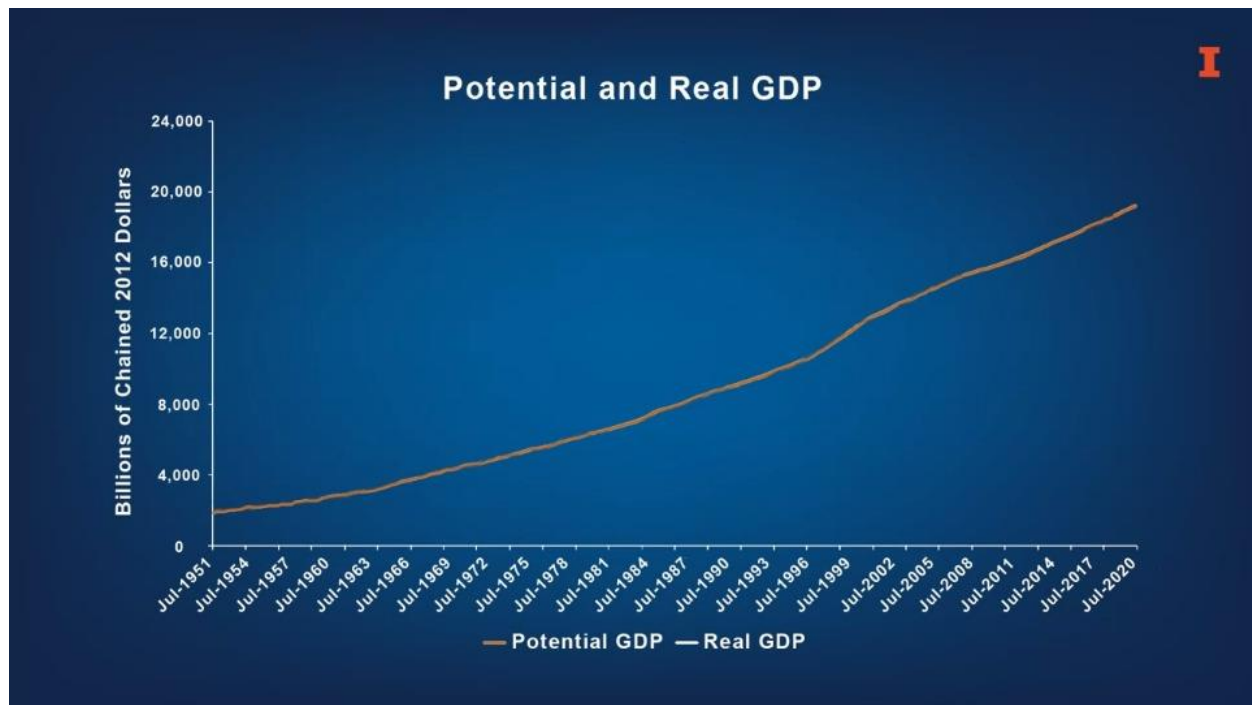
Okun's Law: unemployment $\Leftrightarrow$ GDP

Maximum employment $\Leftrightarrow$ "potential GDP"

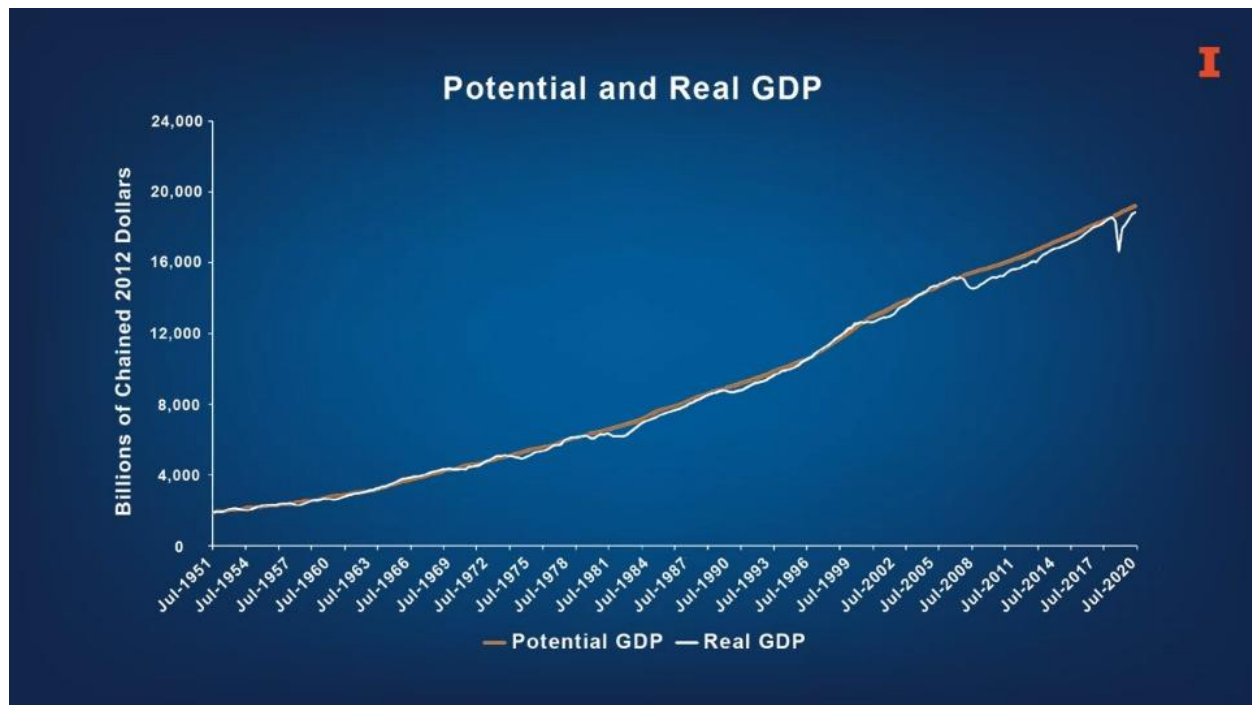
Target: current GDP = potential GDP



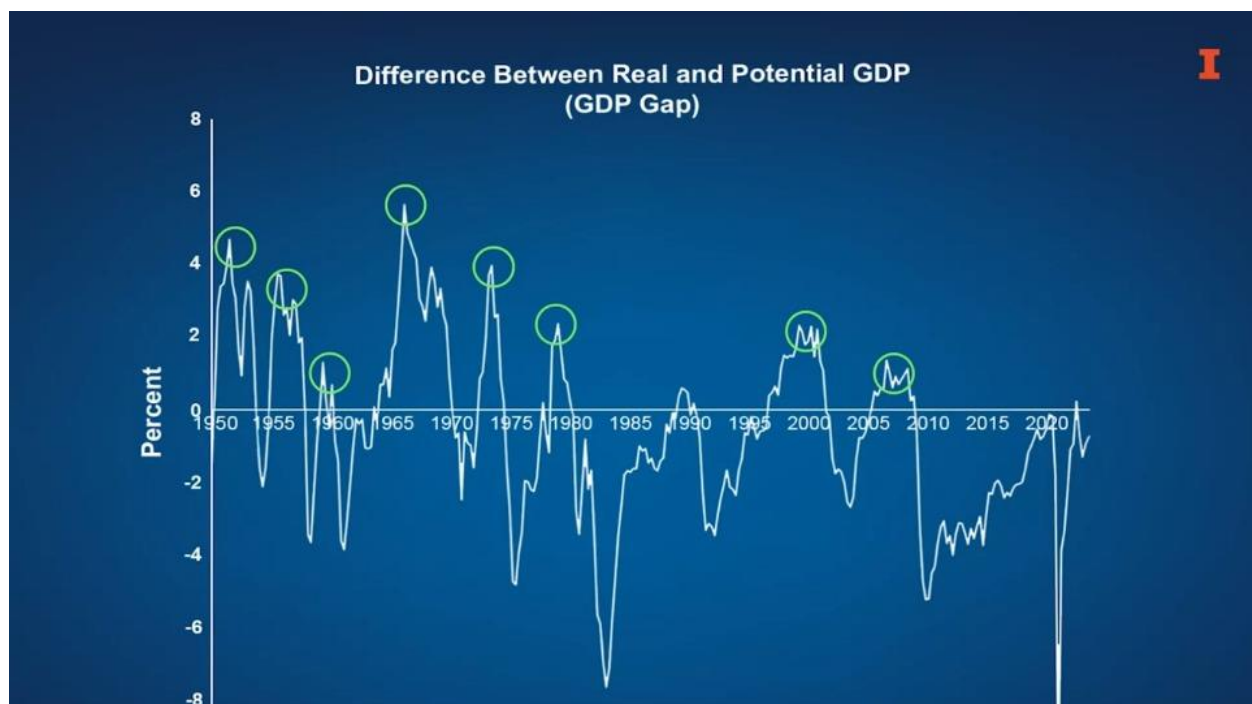
Remember the Federal Reserve dual mandate, maximum employment, and stable prices. Let's first focus on maximum employment and describe how this target can be achieved. Unemployment and GDP are closely connected. This relationship is called Okun's Law. As a result, we often talk about economic growth and unemployment interchangeably. When an economy is fully utilizing its resources, we say that the economy is producing its potential GDP. When the economy is at potential GDP, this implies full or maximum employment. This is just about the best we can hope for. To achieve maximum employment, central bankers need to know if the economy close to potential GDP.



What does this look like in the data? In the chart, you can see an estimate of potential real GDP of the US through 2030. A branch of the US government, The Congressional Budget Office estimates potential GDP. This is a purely theoretical construct. It grows over time due to projected improvements in technology, education, and demographic trends.

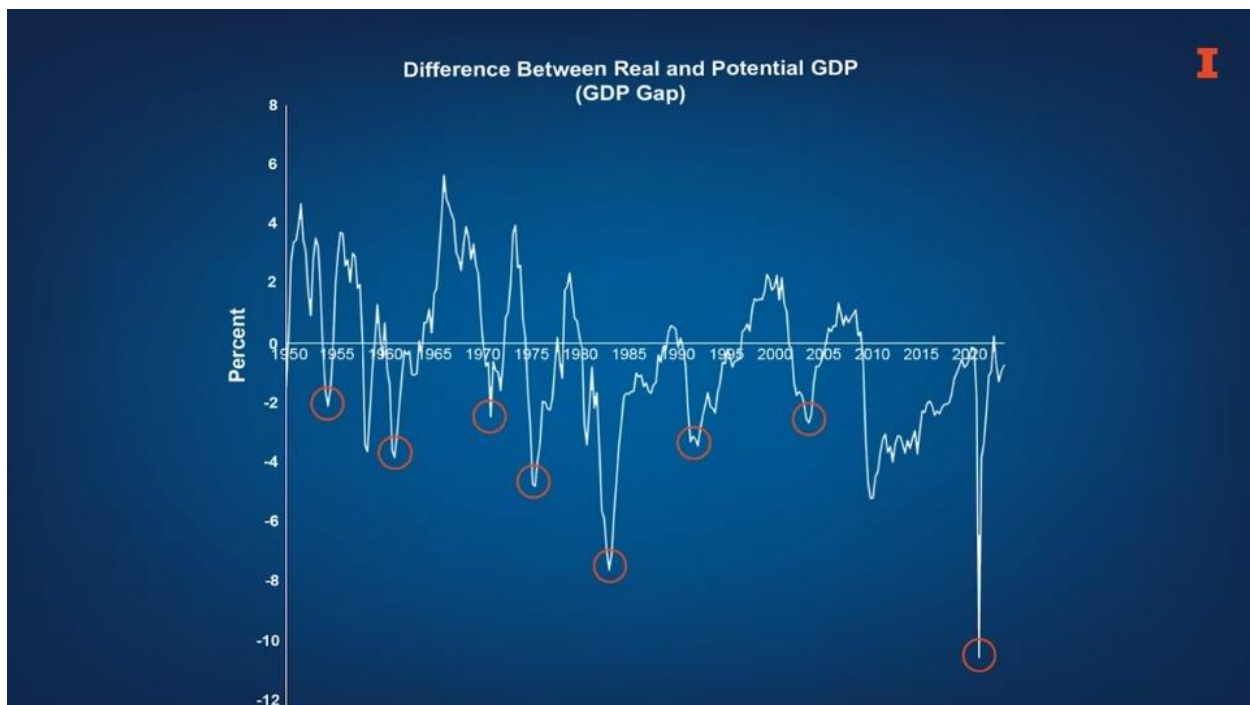


Here's the actual Real GDP. While the two track each other closely, there are notable deviations.

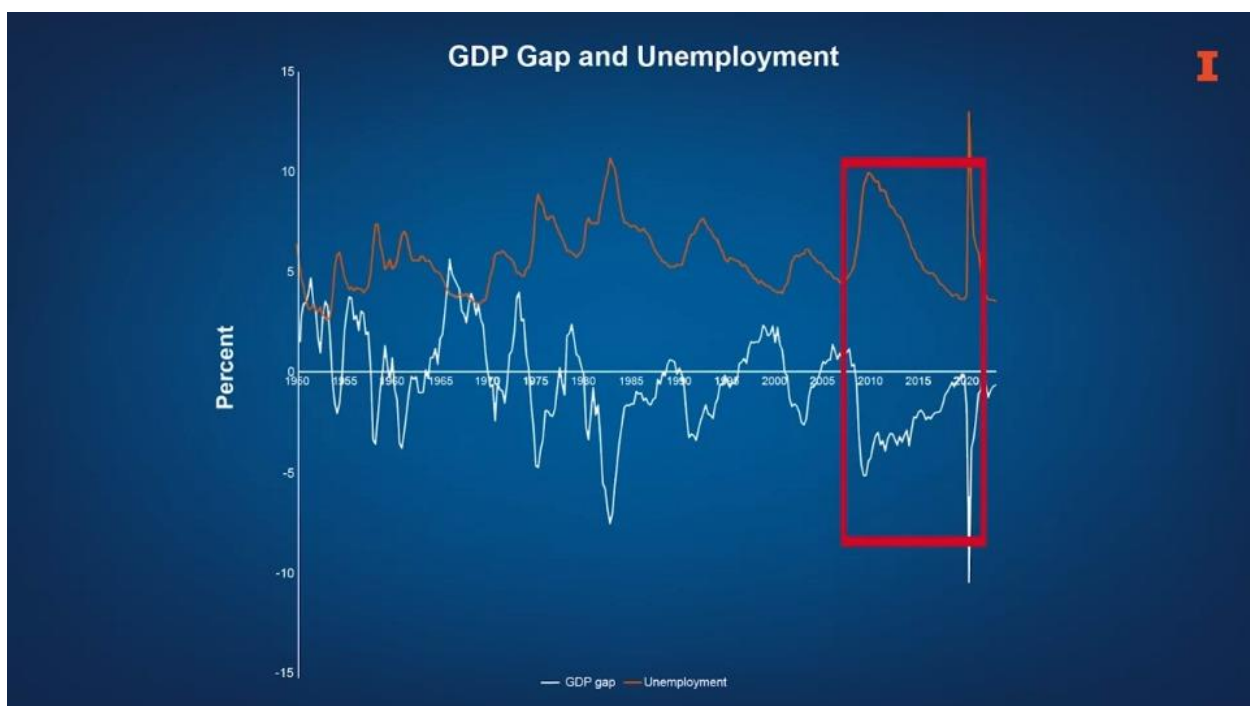


This chart shows these deviations. It is calculated as the percent point difference between the actual real GDP, and potential GDP. This is called the output gap or the GDP gap. There are times when the output gap is positive. These are boom times when the economy runs hot. Resources are used more than they will be in the long run.

Workers may work overtime and extra shifts, and so on. Positive gaps are less common and small in magnitude.

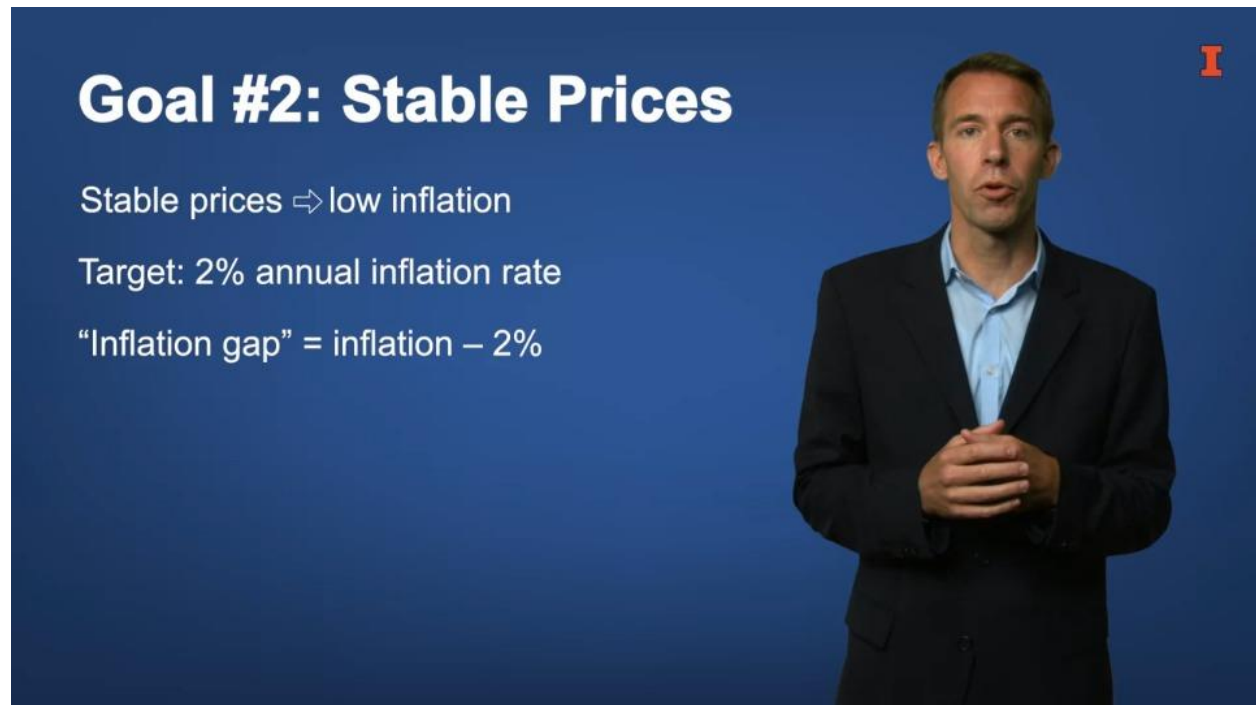


In contrast, negative output gaps can be large and longer lasting.



Let's now layer on the unemployment rate. We see that when the output gap is negative, unemployment is higher than usual. The decade beginning in 2008 is a clear

example. What is the key insight here? Using the output gap as input for monetary policy makes a lot of sense when maximum employment is one of your goals.



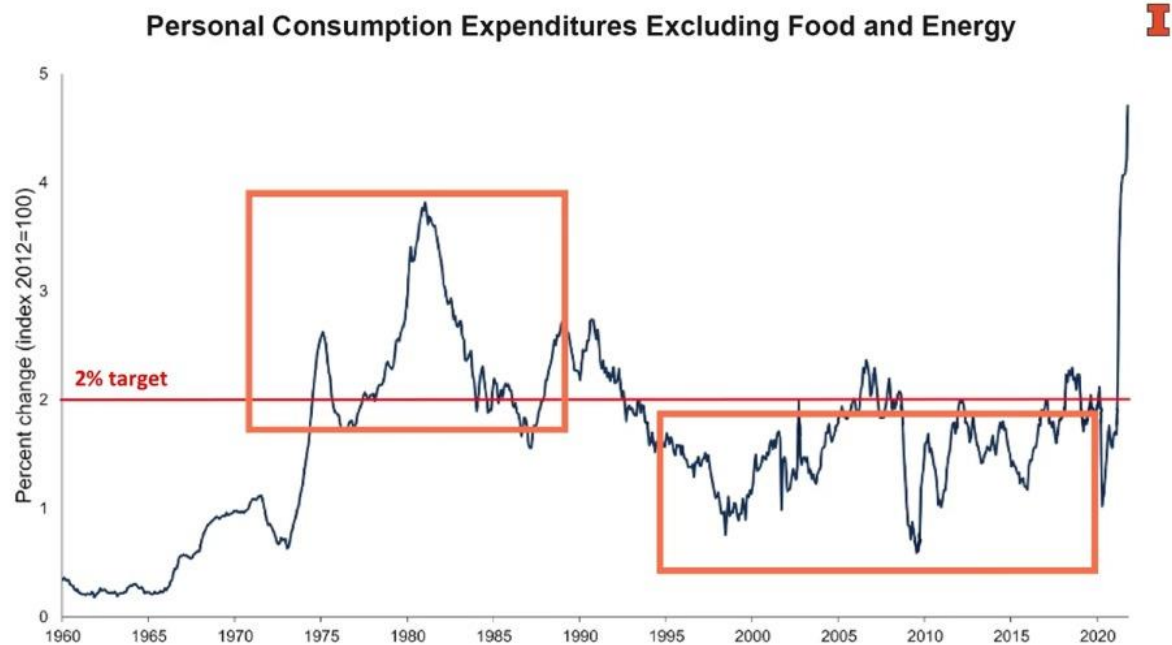
Goal #2: Stable Prices

Stable prices $\Rightarrow$ low inflation

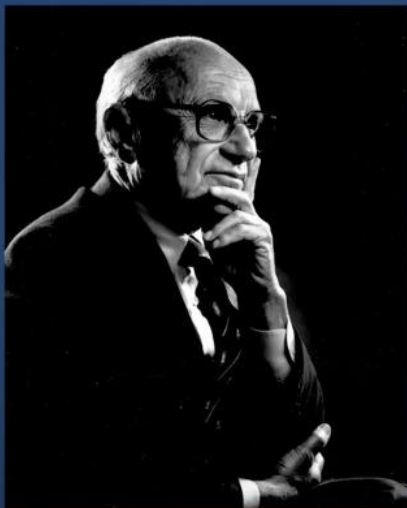
Target: 2% annual inflation rate

"Inflation gap" = inflation – 2%

Moving on, the second part of the Federal Reserve's mandate is stable prices. Stable prices mean a low inflation rate. But how low? Academics such as John Taylor have suggested that Central Banks should use a 2% target. The Federal Reserve seems to agree. Since 2012, the FMC has announced it structures an annual rate of 2% consumer price inflation to be most consistent over the longer run with the Federal Reserve's statutory mandate. Similarly, the European Central Bank shoots for inflation below, but close to 2%. Just with the output gap, we can calculate an inflation gap. This measures how much inflation differs from the Central Bank's 2% target rate.



This graph shows you the annual growth rate of personal consumption expenditure. This is a measure of consumer price inflation. We can see that inflation was above the 2% target in the 1970s and 1980s. On the other hand, from the mid 1990s through 2020, inflation was consistently below 2%.

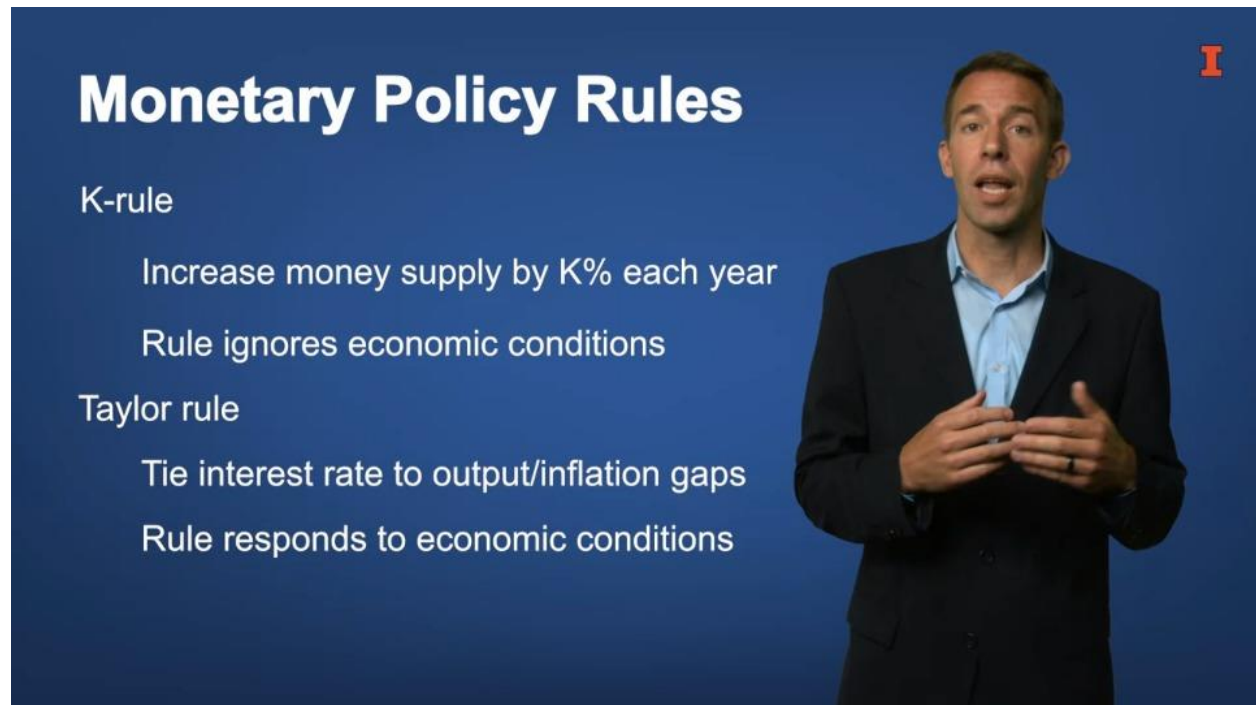


Milton Friedman



How should the Central Bank respond to fluctuation in unemployment and inflation?

Should they respond at all? Nobel Prize winning economist Milton Friedman was skeptical that central banks could make a difference.



Monetary Policy Rules

K-rule

- Increase money supply by K% each year
- Rule ignores economic conditions

Taylor rule

- Tie interest rate to output/inflation gaps
- Rule responds to economic conditions

Instead, Friedman proposed what is now known as the K-rule. The central bank should simply increase the money supply by a constant K percent every year, say 3%. Clearly, this rule ignores economic conditions. The central bank has no leeway and just follows the preprogrammed algorithm. Why is this a good rule? Well, if large increases in the money supply cause inflation, then moderate money growth should keep prices stable. Another advantage of this rule is that financial markets and businesses can perfectly anticipate all monetary policy actions and base their decisions on it. The belief is that keeping the money supply under control and predictable would promote economic and financial stability. The other class of monetary policy rules are designed to respond to economic conditions. If business cycles are costly, then smoothing them out might benefit society. The most famous rule is the so-called Taylor rule, which ties interest rate decisions to fluctuations in output and inflation.


$$\begin{aligned}\text{Nominal Interest Rate (i)} &= \text{Real Interest Rate (r}^*) + \text{Inflation (\pi}^*) \\ &= \quad \quad 0.5\% \quad \quad + \quad \quad 2\% \\ &= 2.5\%\end{aligned}$$

Let's look at the Taylor rule. It was originally proposed by academic John Taylor in the 1990s. To begin, recall that the nominal interest rate is equal to the real interest rate, R , and inflation π . Let's assume that inflation is at the target of 2%. Let's also assume that the real interest rate in equilibrium is 0.5%. Assuming that inflation is sitting at target, the nominal interest rate will be 2.5%. That's what the market will deliver without any intervention by the central bank. However, the central bank can choose a different nominal interest rate if it wants to.

Taylor Rule

$$\text{Target Interest Rate (i)} = 2.5\% + 1.5 \text{ Inflation Gap} + 0.5 \text{ Output Gap}$$

In fact, let's assume that the central bank chooses the target interest rate to explicitly account for the inflation gap and output gaps with weights Alpha and Beta. There's your simple Taylor rule. This says the target interest rate should be whatever the markets want plus an adjustment for economic conditions.

Taylor Rule

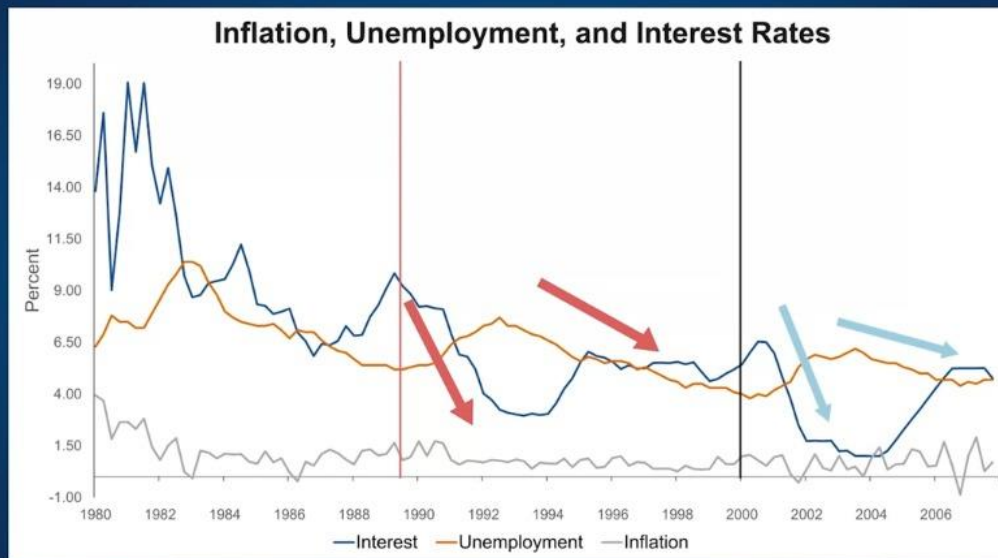
$$\text{Target Interest Rate (i)} = 2.5\% + \alpha \text{ Inflation Gap} + \beta \text{ Output Gap}$$

More precisely, it says that the central bank should increase the interest rate when there is a positive inflation gap or a positive output gap. This should cool down the economy.

Taylor Rule

$$\text{Target Interest Rate (i)} = 2.5\% + \alpha \text{ Inflation Gap} + \beta \text{ Output Gap}$$


And it says the opposite, when there's a negative inflation gap or a negative output gap. This should stimulate the economy. The simple rule captures the elements for decision making at the Federal Reserve. Both stable prices and maximum employment appear in the formula. Of course, we need to find values for the inflation weight Alpha and the output gap weight Beta. Taylor's original recommendation was to give equal weight, setting them both to 0.5. Economic researchers have since conducted numerous studies to see what these weights should be. Nowadays, the consensus for the inflation weight Alpha is 1.5, while the estimate for the output gap weight Beta varies between 0.5 and 1. Using monetary policy to stabilize business cycles has been the norm for many central banks in the modern era, but has this approach been successful? Based on the US experience since the 1980s, you might cautiously conclude, yes.



Here is inflation, unemployment, and interest rates in the United States since 1980. When the 1990 recession started, the Federal Reserve cut the interest rate. Unemployment rates dropped below the prerecession level within four years, and inflation remained low throughout the 1990s. This pattern repeats itself after the dot-com bubble bursts in early 2000. More generally, the low volatility in the macroeconomic variables since the 1990s has been attributed to central bankers and their interest rate decisions.

Summary

Design of monetary policy rules

Monetary policy targets in terms of “gaps”

Taylor rule: Interest rate adjusts
to economic conditions



Let's wrap up. In this session, we covered the design of monetary policy rules. We first described how monetary policy targets could be measured in terms of output and inflation gaps. We then discussed how monetary policy rules can be designed to react to these targets. We saw this in the context of the simple Taylor rule. Finally, we saw how interventions by the Federal Reserve seems to have succeeded in the 1980s and the early 2000s.

Lesson 3-2: Transmission of Monetary Policy

Lesson 3-2.1. How Do Changes in Monetary Policy Affect the Economy?



Learning Objectives

Impact of policy rate changes on:

1. Other interest rates
2. Financial markets
3. Households and businesses
4. Macroeconomy

Hello, and welcome to this lecture on the Transmission of Monetary Policy to the Economy. In this session, we are going to summarize how monetary policy decisions spread through the economy. In the United States, the Federal Reserve adjusts monetary policy in response to changing economic conditions. Remember, the goal is maximum employment and stable prices. The most common response is raising or lowering the short-term interest rate, the federal funds rate. A change in the federal funds rate will lead to changes in other interest rates and in broader financial market conditions. In turn, those changes will then affect the spending decisions of individual households and businesses. Finally, these individual effects will aggregate to the macroeconomic scale. This should feed into economy wide growth, employment, and inflation. Our goal is to understand how this chain of events plays out. This will improve our understanding of how monetary policy works. Let's get started.

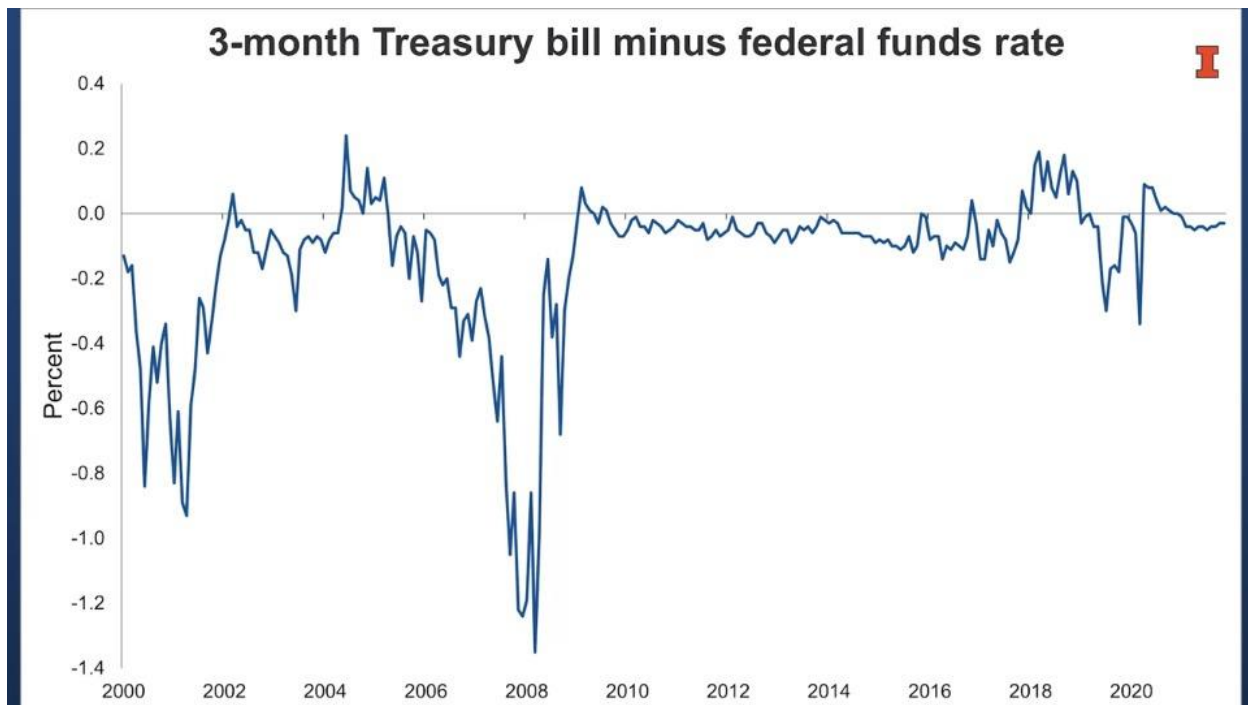
“... **monetary policy** ... has one important advantage—namely **I**
that it gets in all of the cracks.

The one thing that a commercial bank, a broker-dealer, an offshore hedge fund, and a special purpose asset-backed commercial paper vehicle have in common is that they all face the same set of **market interest rates**.

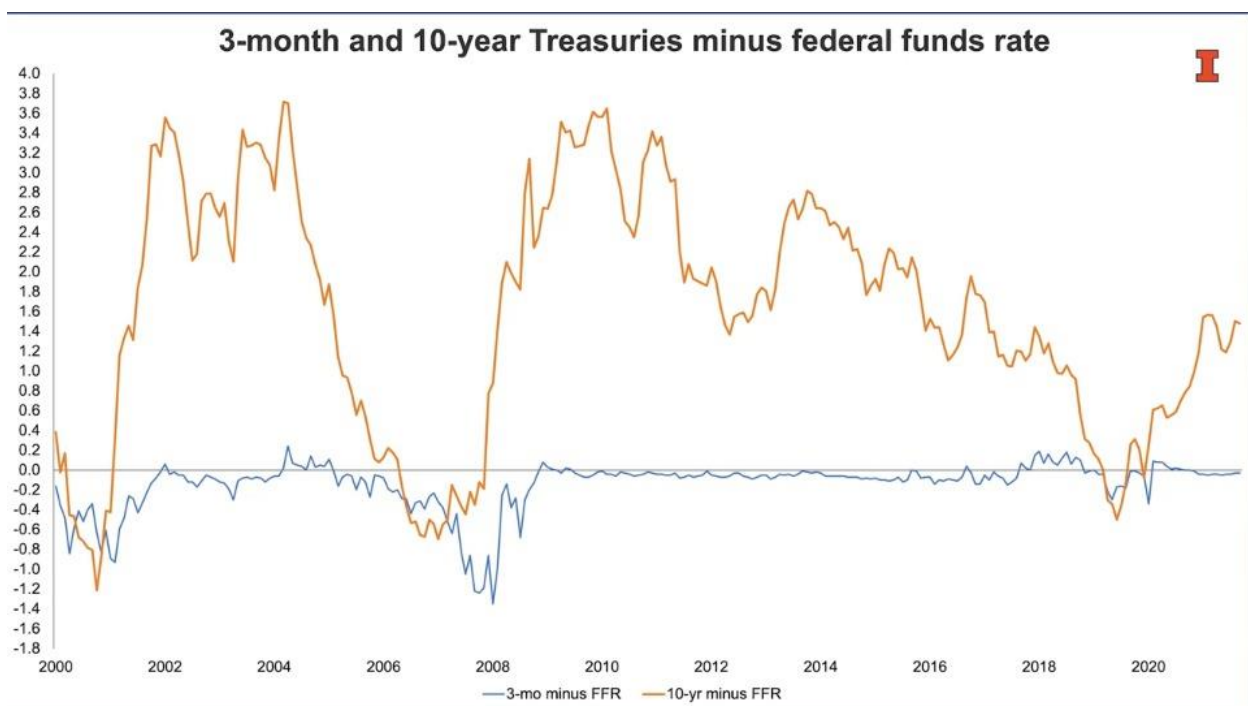
To the extent that market rates exert an influence on risk appetite, or on the incentives to engage in maturity transformation, changes in rates may reach into [all] corners of the market ...”

—Jeremy Stein, Federal Reserve Governor

As the then Federal Reserve Governor Jeremy Stein put it in a speech in 2013, monetary policy has one important advantage, namely that it gets in all the cracks. The one thing, a commercial bank, a broker dealer, an offshore hedge fund, and a special purpose asset backed commercial paper vehicle have in common is that they all face the same set of market interest rates. To the extent that market rates exert an influence on risk appetite or on the incentives to engage in maturity transformation, changes in rates may reach into all corners of the market. Let's first examine the impact of monetary policy on market interest rates. The federal funds rate is an overnight interest rate. Banks can earn this interest rate if they want to. Other short term interest rates are substitute investments. This means that other short term interest rates should track changes in the federal funds rate.



Let's compare the yield. That is the rate of return on three months Treasury Bills and the federal funds rate. The graph shows that the difference between the two is close to zero. This means that the yield on three months Treasury bills moves almost one to one with the federal funds rate.



How about longer-term interest rates? The graph now shows you the difference

between the yield on the 10-year Treasury note and the federal funds rate. You can see that there is no one to one relationship.

Transmission to Market Rates

- Change in federal funds rate (FFR)
- Other short-term rates: yes
- Long-term rates: less so
 - Depends on expectations
 - Forward guidance matters

A man in a dark suit and light blue shirt is standing against a blue background, gesturing with his hands as if speaking. A small red 'I' logo is visible in the top right corner of the slide.

More generally, the effect of a federal funds rate change on short term interest rates will be large. However, the effect on longer term rates will be muted. Long term rates incorporate expectations over longer horizons. These rates respond to how people expect the federal funds rate to change in the future, not just today. For example, suppose markets believe that monetary policy is going to tighten substantially over the next several years. In this case, longer term interest rates today could be much higher than short term interest rates. This is one reason why forward-looking statements about future federal fund rate changes are heavily scrutinized by financial market participants.



Moving on, how does a change in interest rates impact households and businesses? Let's assume that long term interest rates are declining because of an easing of monetary policy. This will make it cheaper for firms to take out loans and invest. Similarly, consumer loans and mortgages become cheaper. This stimulates spending on durable goods such as appliances, as well as homeownership. Lower longer term interest rates will also make yields on US dollar assets less attractive to international investors.



These investors may reduce their demand for dollar denominated assets, lowering the value of the dollar. A weaker dollar tends to boost US exports as US goods become cheaper abroad. At the same time, imported goods become more expensive, giving US consumers and firms an incentive to purchase domestically produced goods.

Transmission to Households/Businesses

Lower long-term interest rates

Businesses

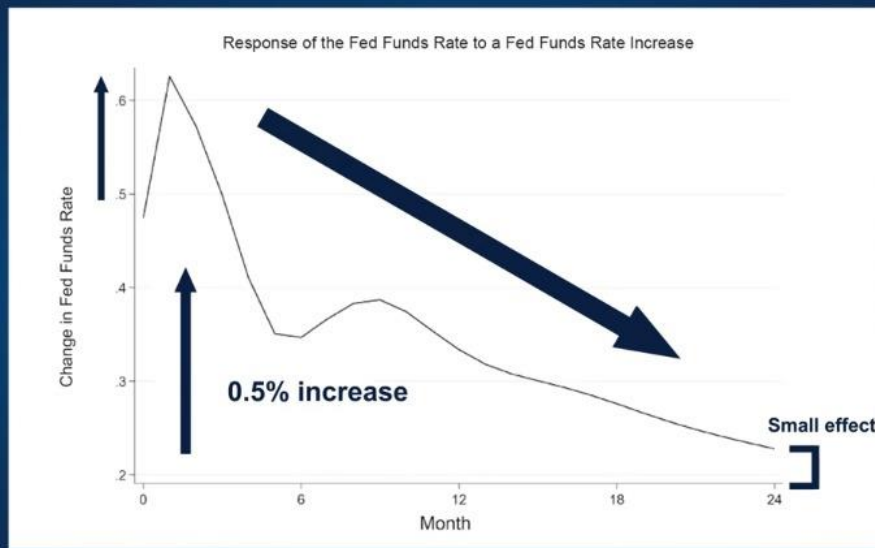
- Lower borrowing costs
- More investment
- Greater export demand
- Greater domestic demand

Households

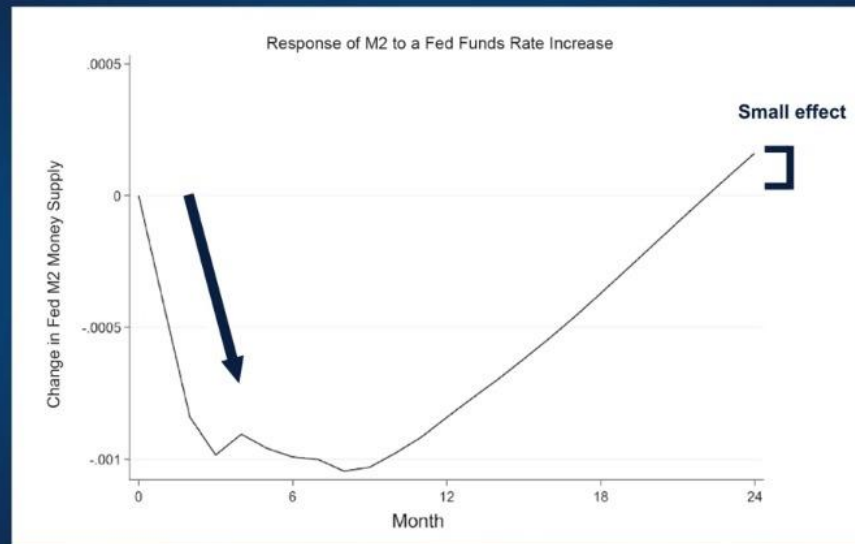
- Lower borrowing costs
- More homeownership
- More consumption

Let's put all these pieces together. We have effects for businesses and effects for households. Longer term interest rate changes also affect stock prices. Usually, we

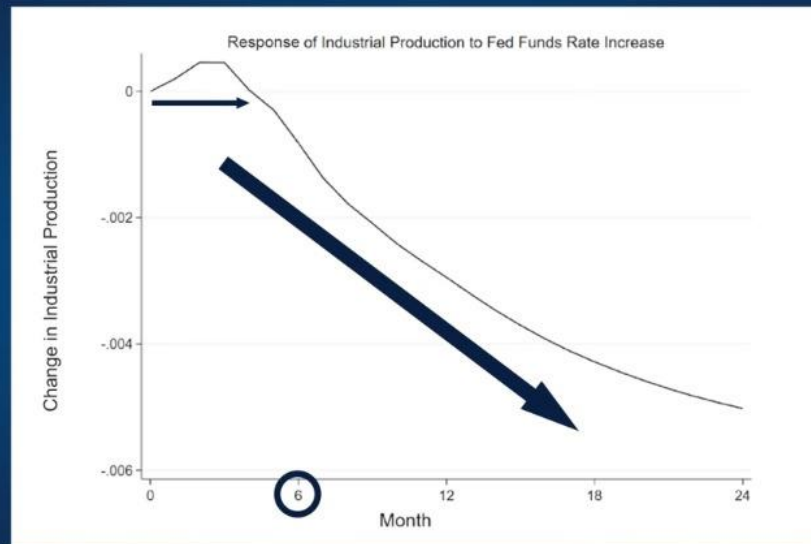
think about the market value of a firm as the discounted value of all future dividends. When longer term interest rates decrease, the discount rate for future dividends becomes more favorable. This increases the market value of a firm and therefore its stock price. This increase in stock prices increases personal wealth, which further increases consumption. The belief is that stimulating the economy with lower interest rates will lead to lower unemployment. This will also lead to higher prices according to the Phillips curve. That's the theory. Let's now see how these effects play out in the data. We will examine a surprise increase in the federal funds rate. We want to know what happens to industrial production, the money supply, the unemployment rate, and inflation. The charts we will see are based on time series statistical analysis estimated on monthly data. We will measure the effects over a two-year time horizon.



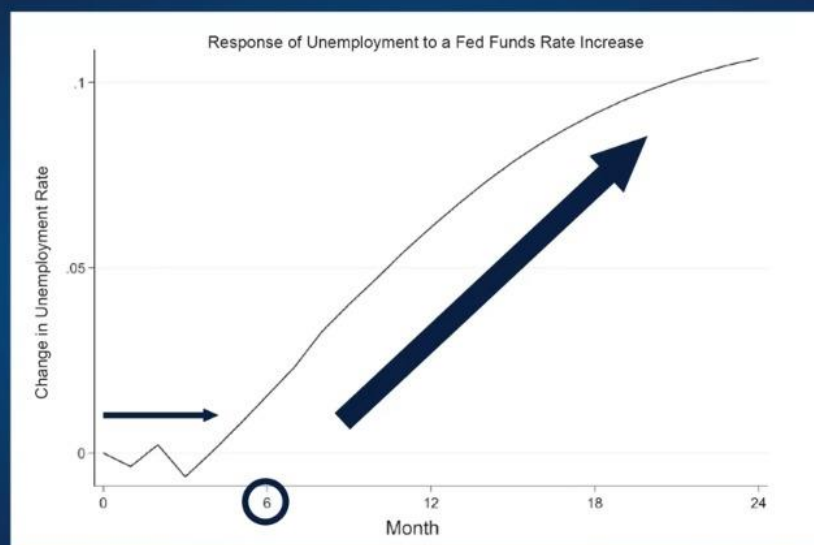
The first graph shows you what happens to the federal funds rate after a one-time surprise increase of 50 basis points. Notice that the interest rate is likely to further increase in the short run. However, over time, the effect of the interest rate surprise wears off and we revert to where we began.



Here is the M2 money supply. The Federal Reserve implements federal fund rate increases with open market operations. Remember that an open market operation involves the Fed selling securities and receiving cash in form of bank reserves in return. Reserves are part of the money supply here measured with M2. Increasing the federal funds rate causes the money supply to shrink. It takes about two years for this effect to subside.



Here's industrial production. The graph shows that it takes about six months for industrial production to go down. This is an important point. Monetary policy affects the economy there is a lack. This happens for at least two reasons. First, firms may have already received funding for investment plans and are still implementing them. Second, it may take some time until changes in the interest rate affect consumer spending.



Unemployment is the mirror image of industrial production. It takes a couple of months,

but with industrial production declining, the unemployment rate starts to rise. In addition, there is evidence that inflation goes down when the federal funds rate is raised. If the Central Bank increases the federal funds rate because it is concerned about the economy running hard, then over the medium run, unemployment will increase, and inflation will come down.

Summary

Monetary policy first affects financial markets
Interest rates and stock market

Then broader economy with a time lag
Consumption, investment,
unemployment, inflation

Statistical analysis confirms these effects



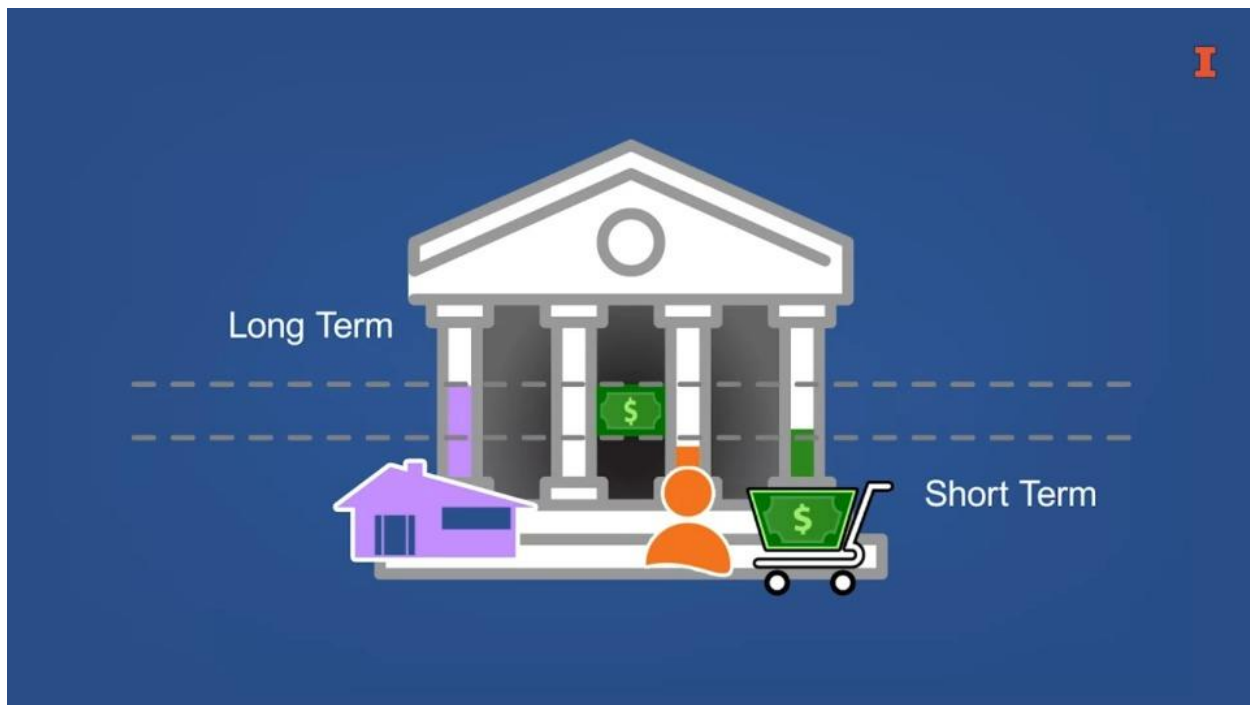
Let's wrap up. In this class, we have summarized how monetary policy transmits to financial markets and the real economy. On aggregate level we described how monetary policy should affect short- and longer-term interest rates, exchange rates, stock prices, and eventually consumption and investment. Then we looked at empirical evidence confirming that after the price increase in the federal funds rate, the money supply falls immediately and then industrial production goes down, unemployment increases, and inflation declines. These effects take time to appear, indicating that monetary policy affects the economy with the time lag.

Lesson 3-2: Transmission of Monetary Policy

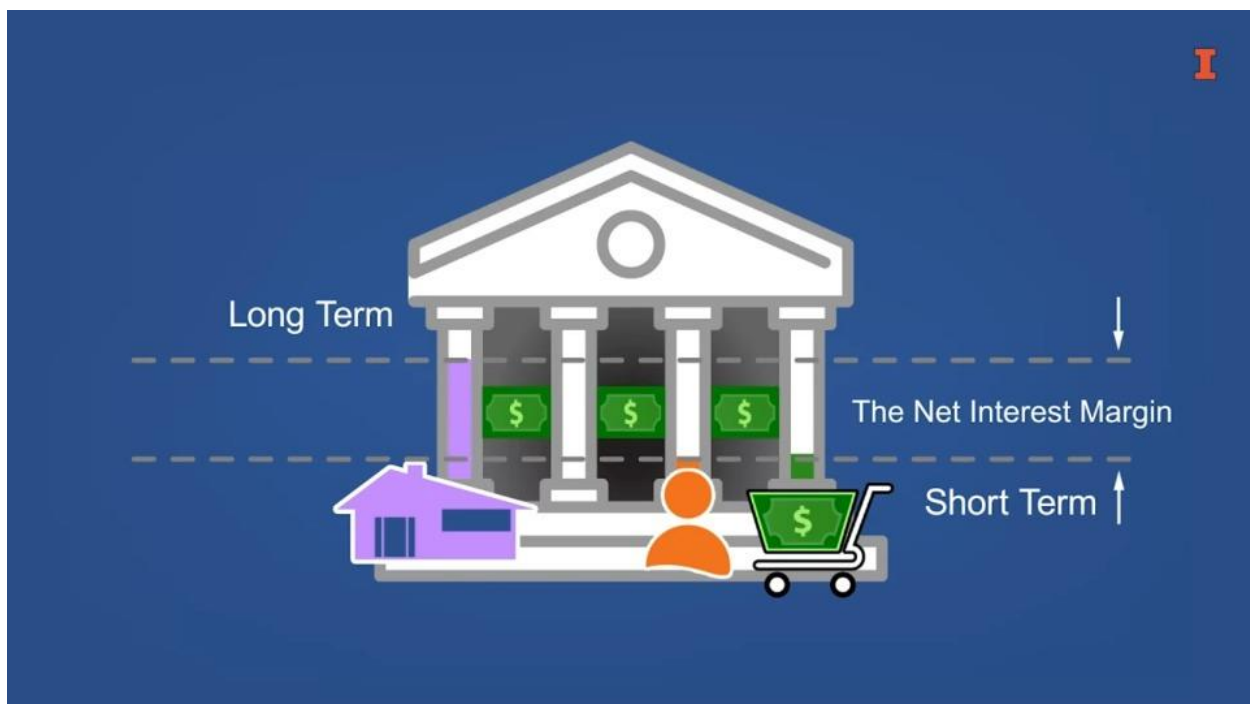
[Lesson 3-2.2. Monetary Policy Transmission Through Banks](#)



Hello and welcome to this lecture on the bank lending channel of monetary policy. In this class, we examine how monetary policy affects credit supply from banks. We will discuss three mechanisms. Net interest rate margin, reserves, and deposits.

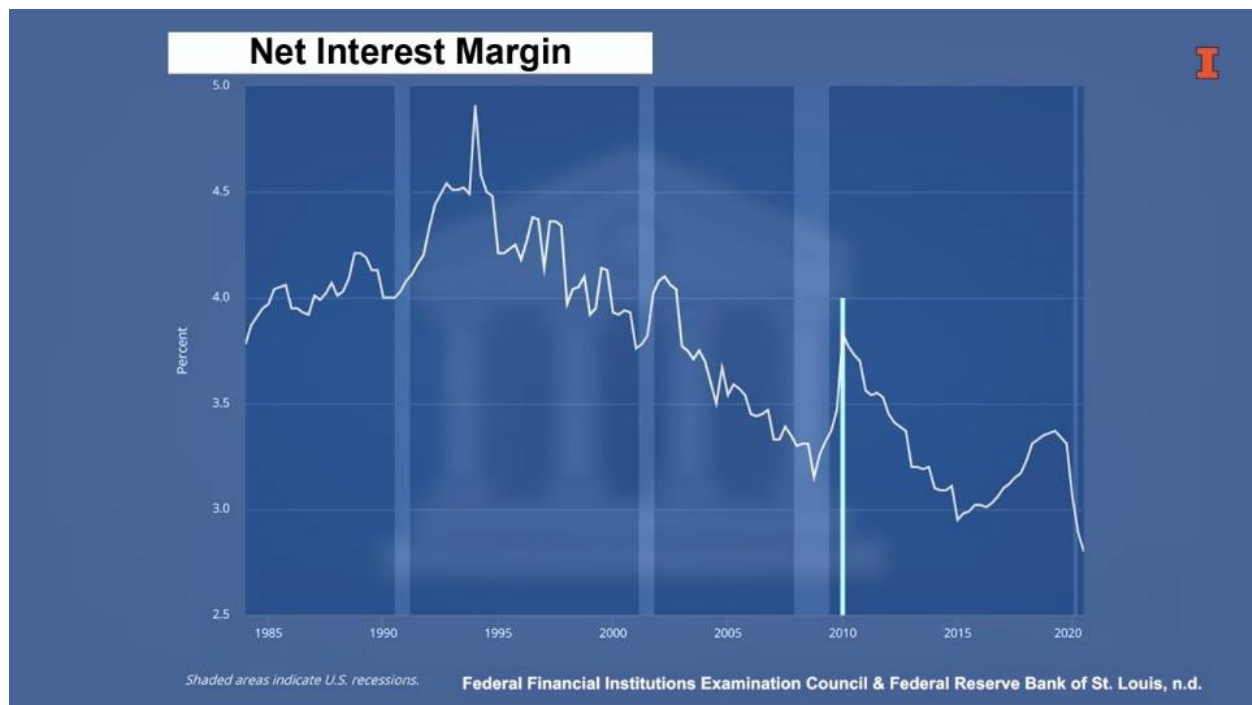


Consider the business model of a bank. A bank borrows short-term from depositors and money market funds and uses these funds to originate longer-term loans. Since short-term interest rates are usually lower than long-term interest rates, the bank makes money by exploiting this difference between short-term and long-term interest rates.

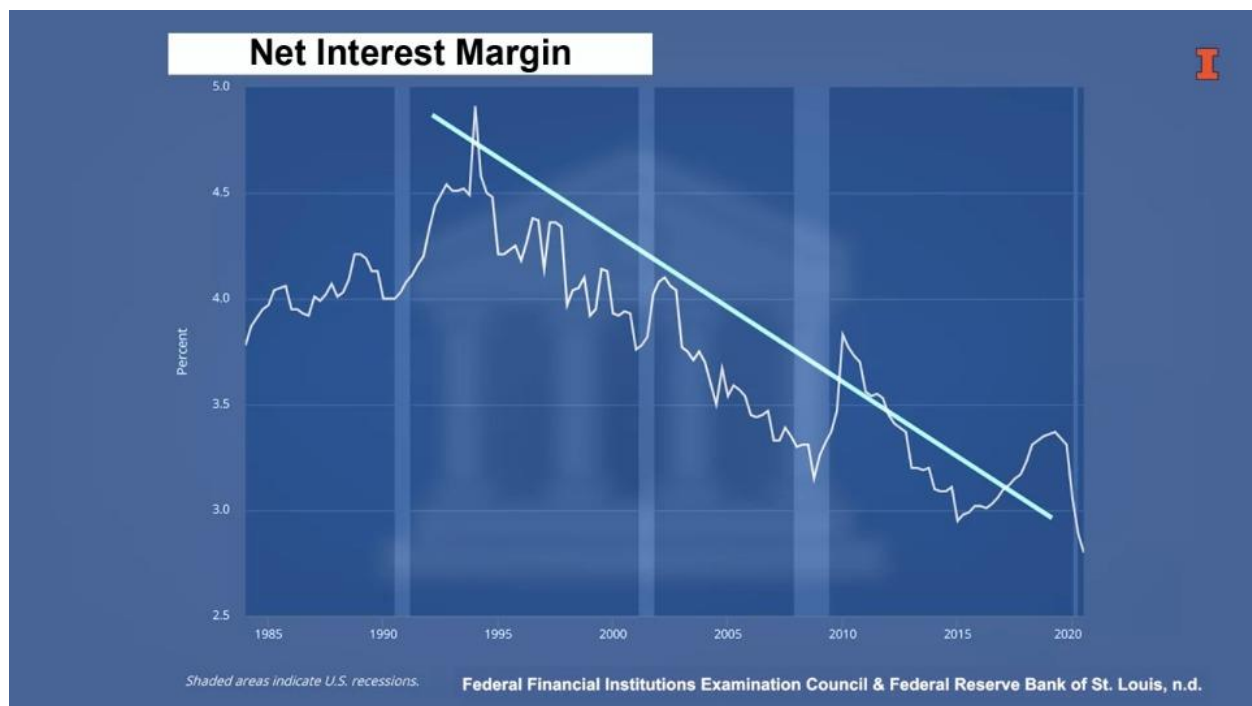


The difference between net interest income of banks from loans and mortgages, and interest expenses paid by the banks to depositors is called the net interest margin. The

net interest margin measures the profitability of a bank. The larger the net interest margin, the higher the profits.

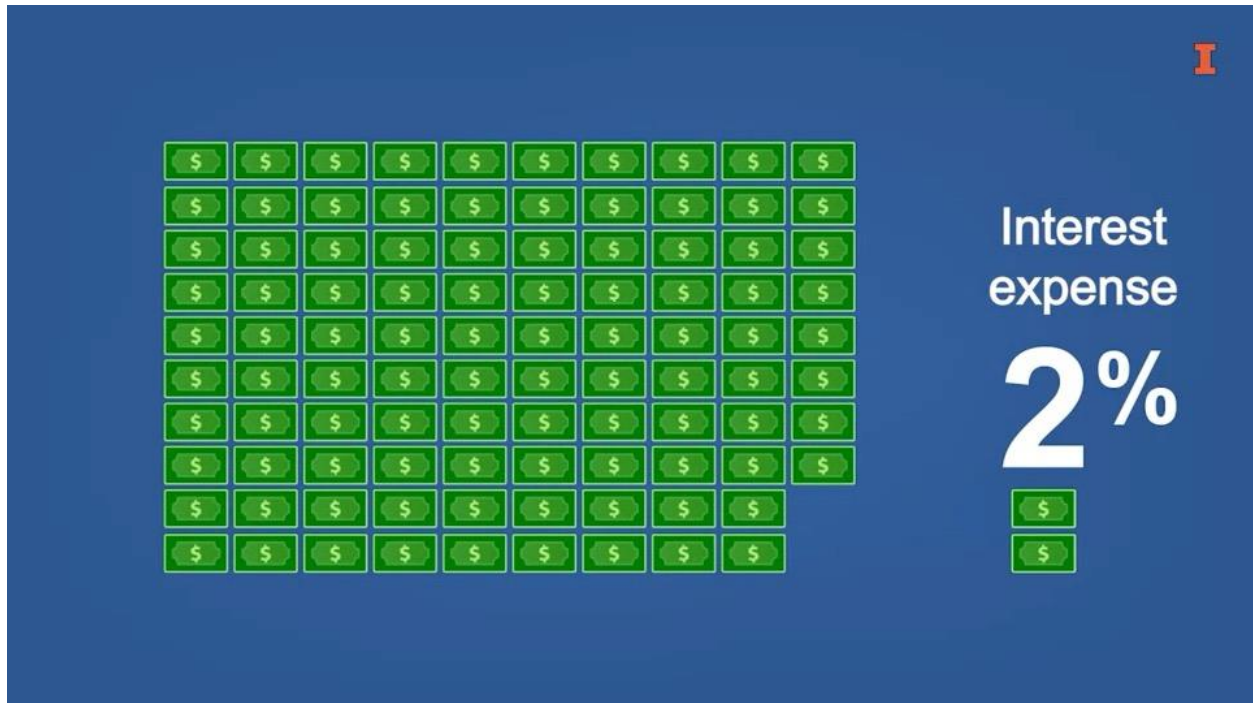


This graph shows you the net interest margins of banks in the United States since the 1980s. As you can see, the net interest margin of banks has decreased over time.

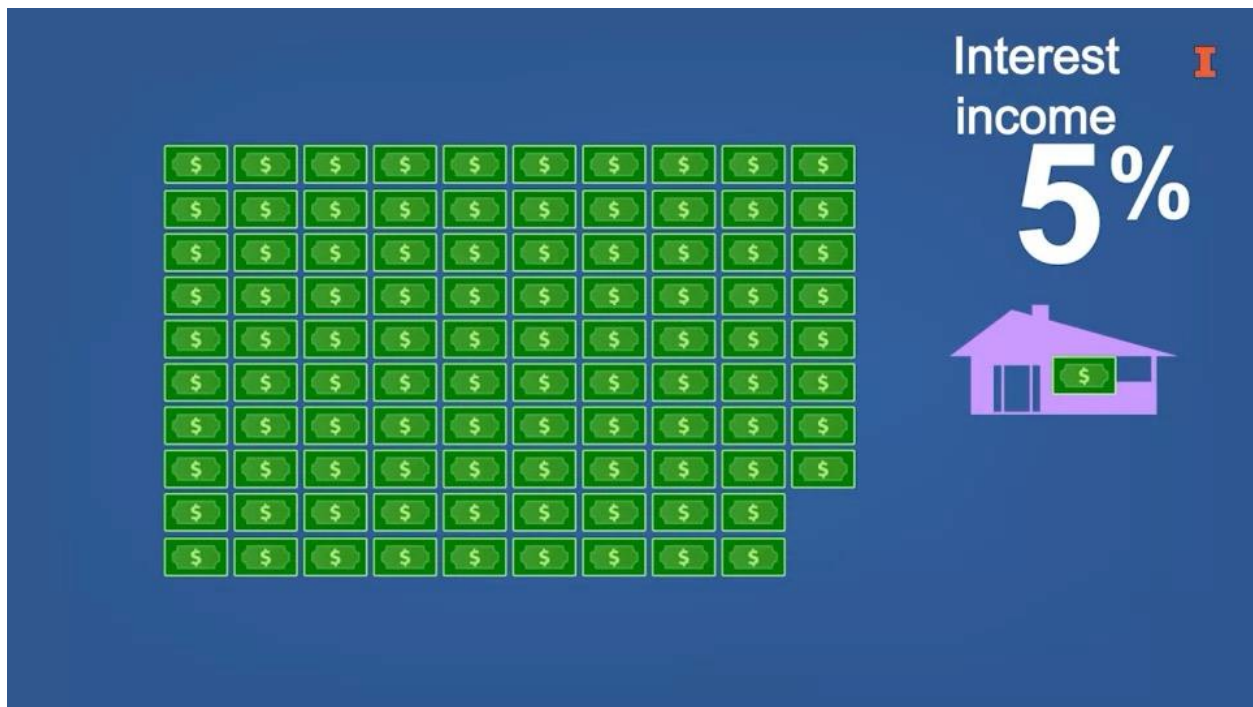


At the same time, you can see that the net interest margin significantly increases

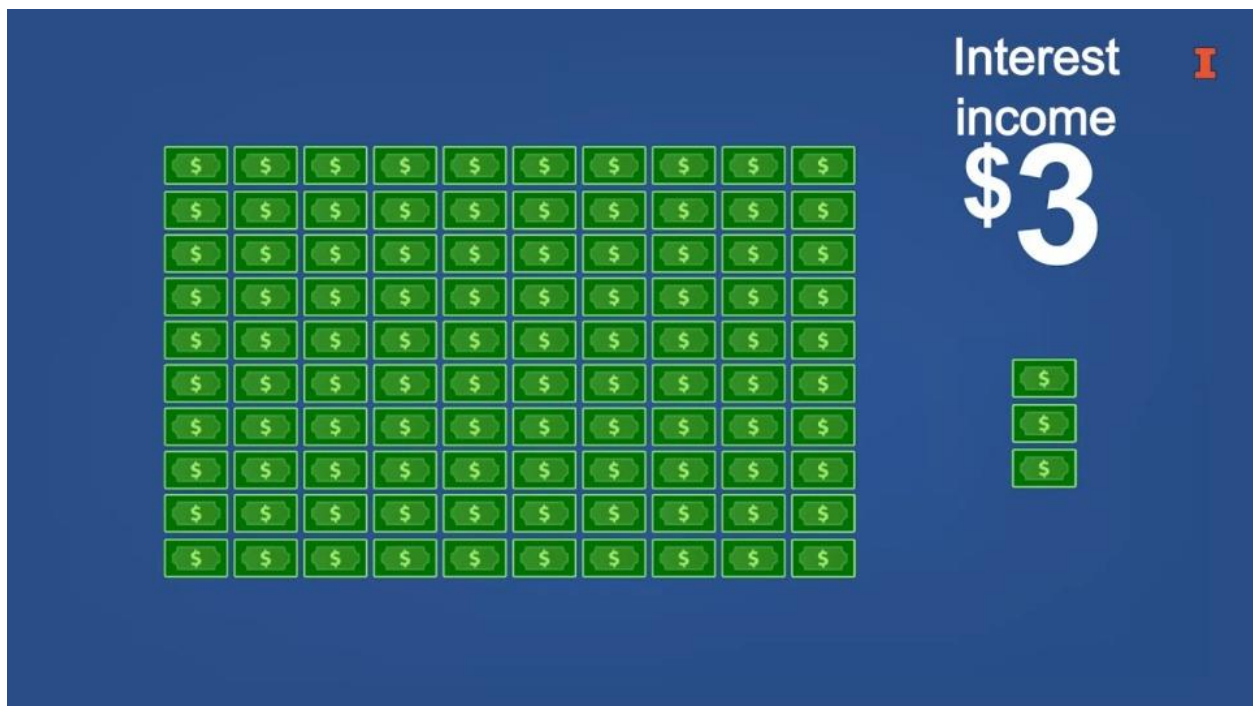
around 2010, when the Federal Reserve cut the federal funds rate to zero for the first time.



Let's see how monetary policy affects the net interest margin, in a simple example, consider a bank that is entirely funded by deposits worth \$100. The interest rate on deposits is 2% so total interest expenses are \$2.



The bank has issued one \$100 fixed-rate mortgage that pays 5% interest.



The interest income is \$5 less \$2 interest expenses gives you a net interest margin of \$3 or 3%.

Policy Rate Increase $\Rightarrow$ Bank Profits I

Fed increases FFR: 2% $\Rightarrow$ 3%

Interest expense (IE): 2% $\Rightarrow$ 3%

Interest income (II): unchanged (5%)

Net interest margin (II – IE): 3% $\Rightarrow$ 2%

Now assume that the Federal Reserve increases the federal funds rate from 2% to 3%. Since the market for federal funds is an overnight lending market, all other overnight and short-term interest rates will increase. Deposits are redeemable on demand, and as such, are very similar to overnight loans. If a bank does not want to lose deposits, the bank will have to increase the deposit rate accordingly. The bank must pay 3% for its deposits, but the interest rate on the fixed-rate mortgage has not changed. The net interest margin, therefore, falls from 3% to 2%.

Channel #1: Net Interest Margin (Profits)

Change in FFR $\Rightarrow$ profits $\Rightarrow$ new loans

Depends on:

- Share of fixed-rate loans

- Rate-sensitive funding



A falling net interest margin means lower profits. Lower profits in turn mean less capital for new loans. In other words, monetary policy directly affects bank profits and therefore credit supply of banks. Of course, how strongly monetary policy impacts profits depends on the share of fixed-rate loans a bank has on its balance sheet. Some loans, such as loans to firms are typically floating interest-rate loans and hence the net interest margin on these loans is not affected by monetary policy.

Channel #2: Bank Reserves

Monetary tightening $\Rightarrow$ FFR up

How?

- Open market operation

- Removes bank reserves

Fewer funds to lend out



The second way monetary policy affects banks is through reserves held by banks in their Federal Reserve accounts, those are called federal funds. When the Federal Reserve raises the federal funds rate, it typically does so by conducting open market operations. That is, the Federal Reserve sells securities to banks. The banks pay for the purchases of securities with reserves. With less reserves available in the federal fund market credit supply in the short term, funding market goes down. Banks use the federal funds market for short-term liquidity needs, for instance, when issuing loans. Of course, banks can also borrow in other markets, but often that funding is more expensive or requires collateral like for instance, the Repo market. When there's less funding available to banks, or the remaining available funding is more expensive, then banks reduce their credit supply. Hence, monetary policy reduces the credit supply of banks by reducing reserves.



How does monetary policy affect banks?

Channel #3: Deposit flows

Another way banks are affected by monetary policy is through deposits. Recall the simple example I gave for the net interest margin. In that example, the banks match the increases in the federal funds rate. However, as most of you know from your checking and savings accounts, banks are usually not fast when it comes to increases in interest rates on checking and savings accounts.

Channel #3: Deposit Flows

Change in FFR

Depositor withdrawals

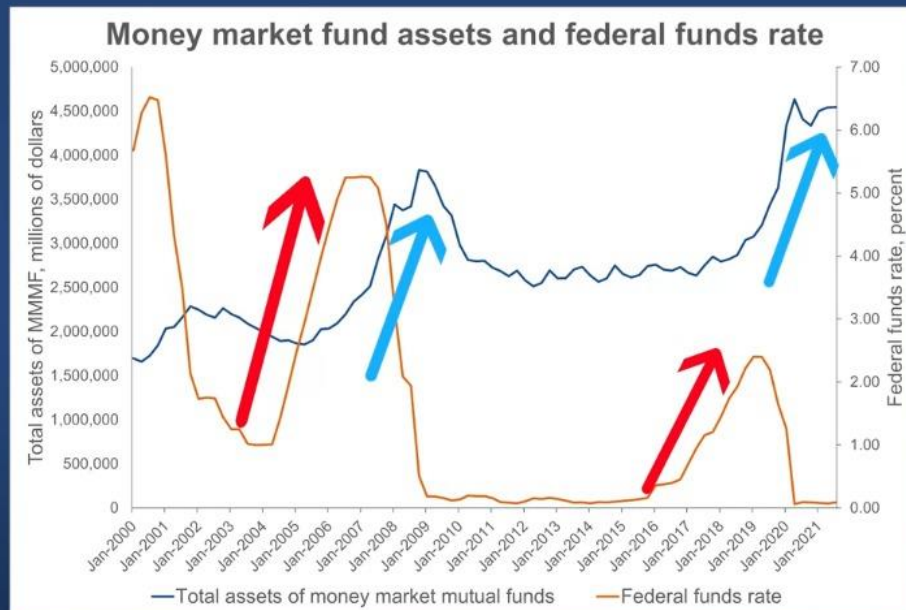
Small deposits “sticky”

Large deposits sensitive



Most depositors tend to not look for investment alternatives because it takes time. It

takes even more time to switch banks because you must change all your bank information everywhere. Even so, depositors have alternatives most find it too costly or too tedious to take advantage of these alternatives. Therefore, retail deposits are referred to as sticky and not sensitive to interest rates. However, some depositors especially those with sizable deposits, do switch. They tend to withdraw their deposits from banks when the difference between the deposit rate and interest rates on alternative investments increases. The closer substitute to deposits is money market mutual funds.



The graph shows total assets of money market mutual funds and the federal funds rate. As you can see, when the Federal Reserve increases the federal funds rate, for instance between 2004 and 2006 or 2016 and 2019 assets of money market mutual funds grew much faster.

Channel #3: Deposit Flows

Change in FFR

Depositor withdrawals

Small deposits “sticky”

Large deposits sensitive

Decisions for bank

Increase rates for all deposits

Reduce credit supply



The reduction in deposits also means that banks have less funding and hence must reduce their credit supply. You may wonder why this behavior is optimal for banks. Banks face a choice between losing some deposits and foregoing some opportunities to originate loans and increasing interest rates on all deposits. For differently, would you rather lose 5% of deposits and originate 5% less loans or increase the interest rates on all deposits by 1% age points and keep your current pace of lending? Recent research suggests that banks are strategic about where they increase deposit rates. Within the United States, deposit rates tend to increase faster in places where there is more bank competition. This suggests that the easier it is to switch banks, the more banks must respond to changes in interest rates to retain their deposits. The flip side of this finding is that in places or countries with highly concentrated banking system, this deposit channel of monetary policy is less important.

Summary

Higher federal funds rate impacts banks:

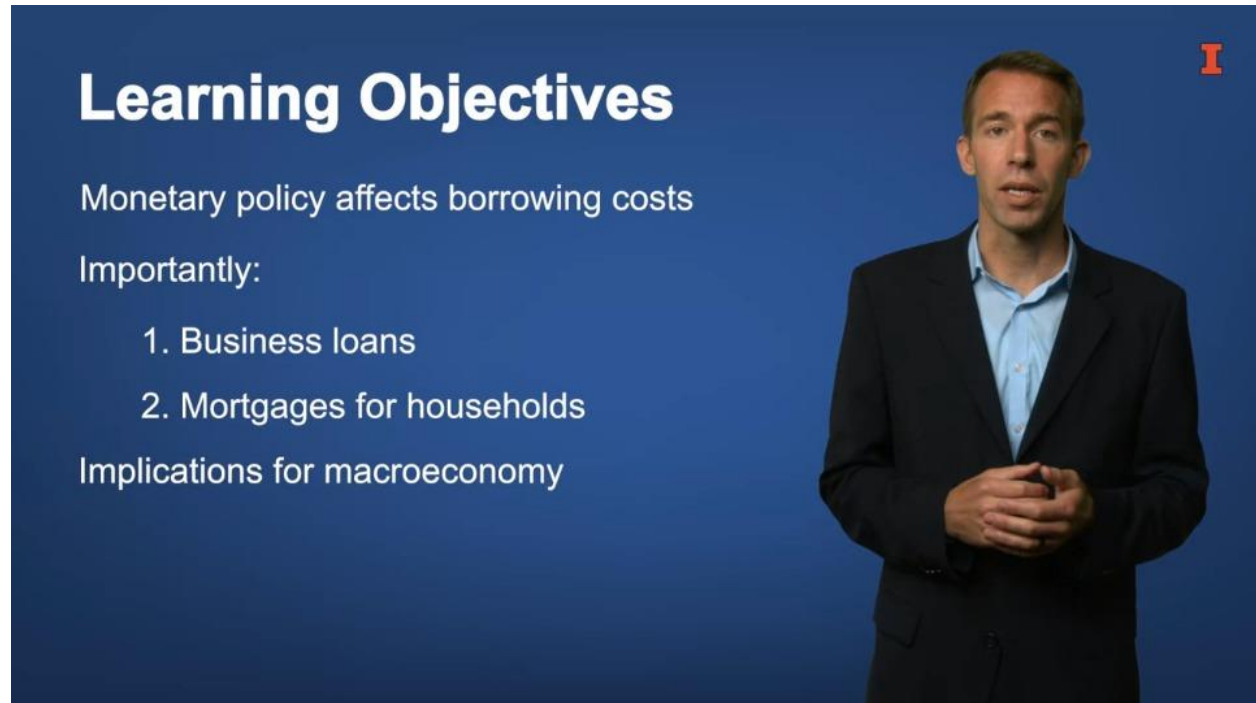
1. Profits may fall
2. Fewer reserves to lend out
3. Fewer deposits to lend to customers



What have we learned in this lesson? Monetary policy affects banks through three channels. First, increases in the federal funds rate compresses the net interest margin reducing profits. Second, increases in the federal funds rate reduce reserves available in the banking system. Third, increases in the federal funds rate led to an outflow of deposits reducing bank funds. All three channels imply that increases in the federal funds rate reduce the overall supply of credit by banks.

Lesson 3-2: Transmission of Monetary Policy

Lesson 3-2.3. Monetary Policy Transmission Through Banks



Learning Objectives

Monetary policy affects borrowing costs

Importantly:

1. Business loans
2. Mortgages for households

Implications for macroeconomy

Hello, and welcome to this lecture on the transmission of monetary policy through businesses and households. Our goal in this class is to better understand how changes in the interest rates are passed through to borrowing costs. We will see how these matters for businesses, particularly small and financially constrained firms that do depend on variable rate bank loans. This can have knock-on effects for investments and jobs in the economy. We will then examine how the same channel impacts households. Here, we will focus on the housing market and mortgage loans. We will see how changes in interest rates matter for households with the option to refinance their mortgage. Then we will discuss the implications for aggregate demand in the economy. Let's get started.

Borrowing by Business Size

Large firms $\Rightarrow$ bonds (fixed-rate debt)

Small firms $\Rightarrow$ bank loans only

Floating rate debt

$\% = \text{benchmark (variable)} + \text{margin (fixed)}$

Benchmark: LIBOR, SOFR,
prime rate, etc.



Let's first look at businesses. Almost all firms take out some form of debt. The largest corporations have access to the bond market, which is fixed rate debt. Smaller firms typically only have access to bank loans. These business loans often have a floating or variable interest rate. This means the loan interest rate is equal to some floating benchmark plus some fixed mark-up. The floating benchmark could be one of in the past, the London Inter-Bank Offer Rate or LIBOR, the Secured Overnight Financing Rate, or SOFR, or possibly the prime rate, which is the interest rate offered to the best borrowers. In general, this means that the interest rate that the firm pays moves with short term interest rates.

Business Loan Example

Loan = \$1M

Loan interest rate = SOFR + 2%

Initially, FFR = SOFR = 4%

Loan rate = 6%

Annual interest = \$60,000



Let's see how this connects to monetary policy with an example. Assume that a firm has one-million-dollar business loan. Let's assume that the interest rate on this loan is SOFR plus 2%. Like the federal funds rate, SOFR is an overnight interest rate and the two move closely together. Let's assume that the federal funds rate and SOFR are currently 4%, this means the loan interest rate is 6% and total annual interest payments are \$60,000.

Business Loan Example

Loan = \$1M

Loan interest rate = SOFR + 2%

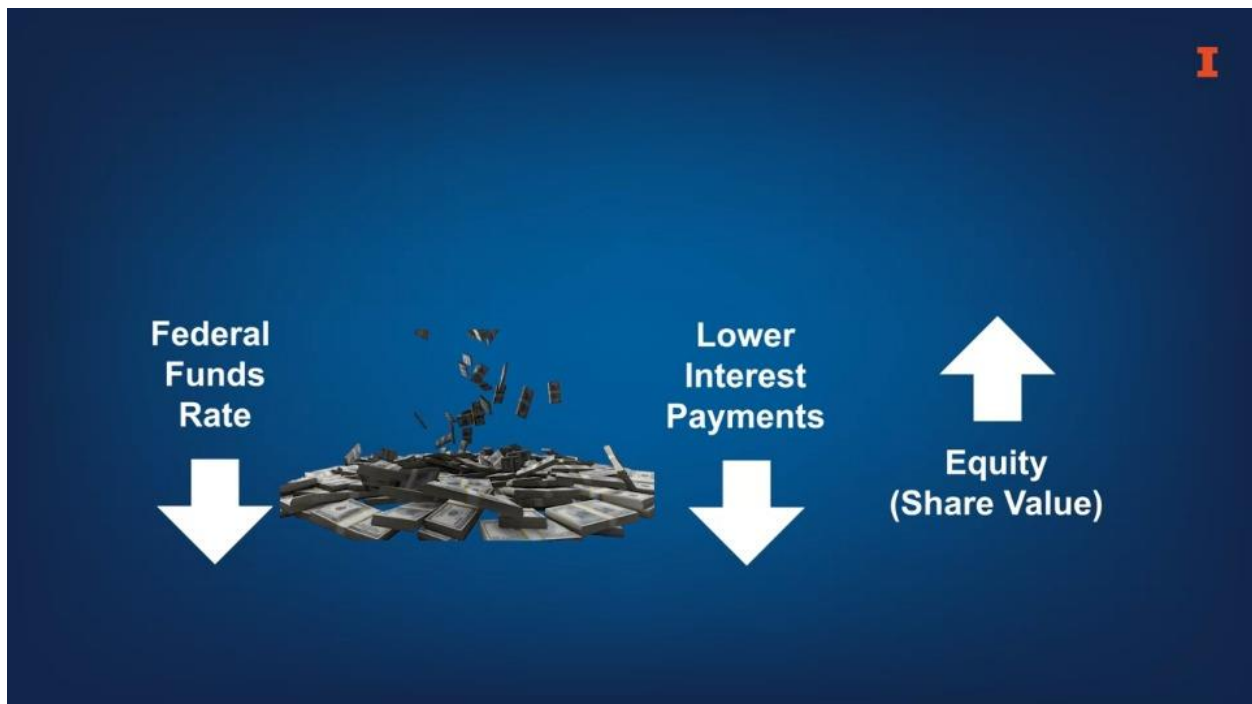
After rate cut, FFR = SOFR = 1%

Loan rate = 3%

Annual interest = \$30,000

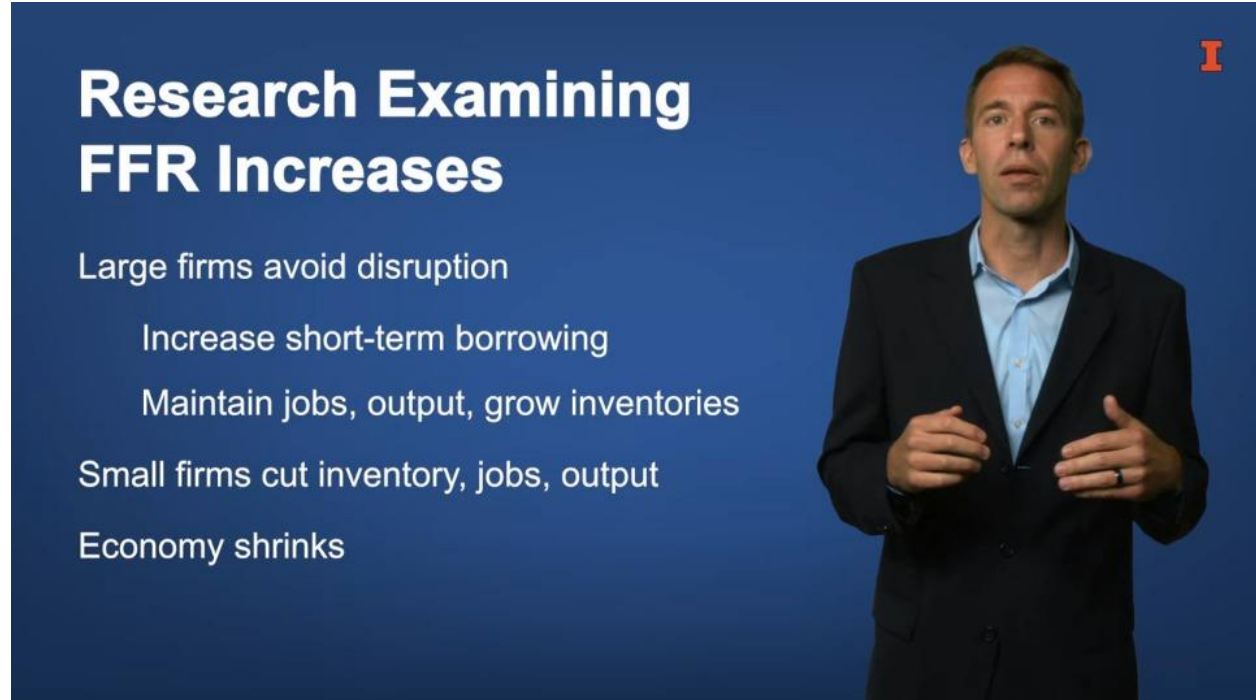


Now let's assume that the FMC decides to cut the federal funds rate to 1%, the SOFR will follow this downward move. This means that the interest rate on the firm's loan will reset from 6% to 3% reducing the interest rate payments to \$30,000. The reduction in the federal funds rate hence frees up \$30,000 that the firm could use for investment and hiring.



More generally, a reduction in the federal funds rate will increase the profits of

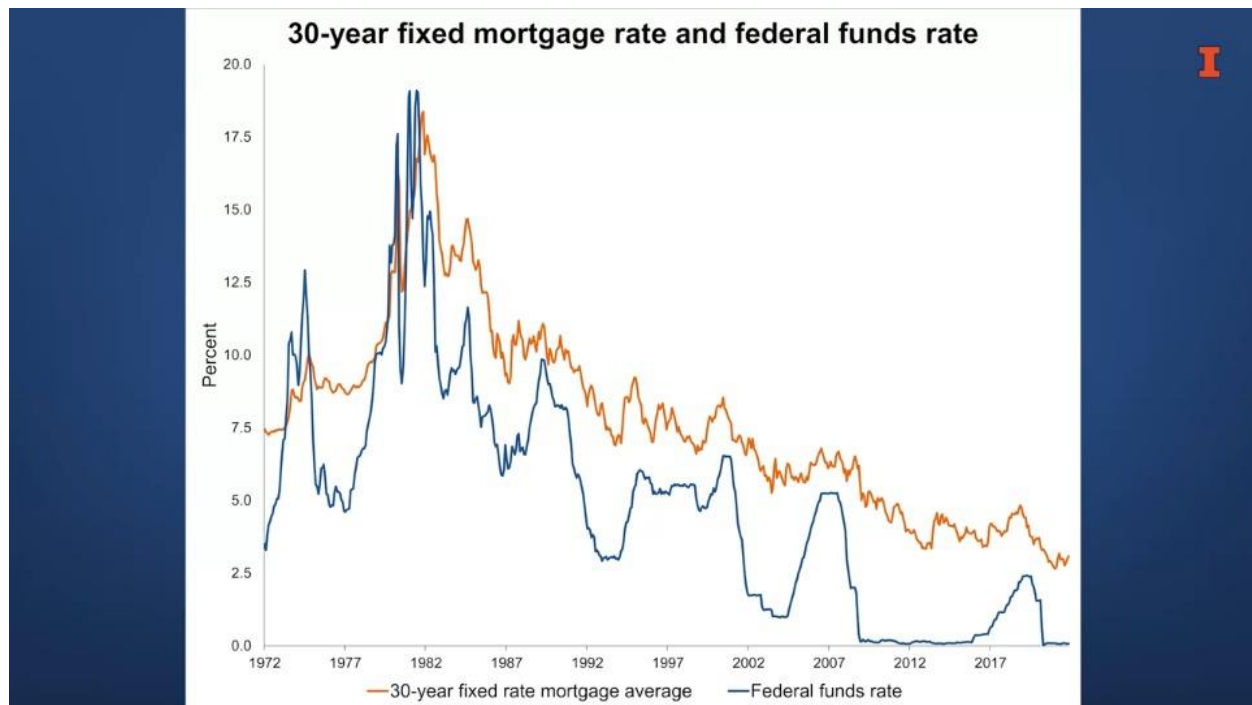
leveraged firms. Another implication of higher profits is that the firm's balance sheets, and equity valuation will improve. With higher profits, the firm is in a better financial position. This may further reduce borrowing costs.



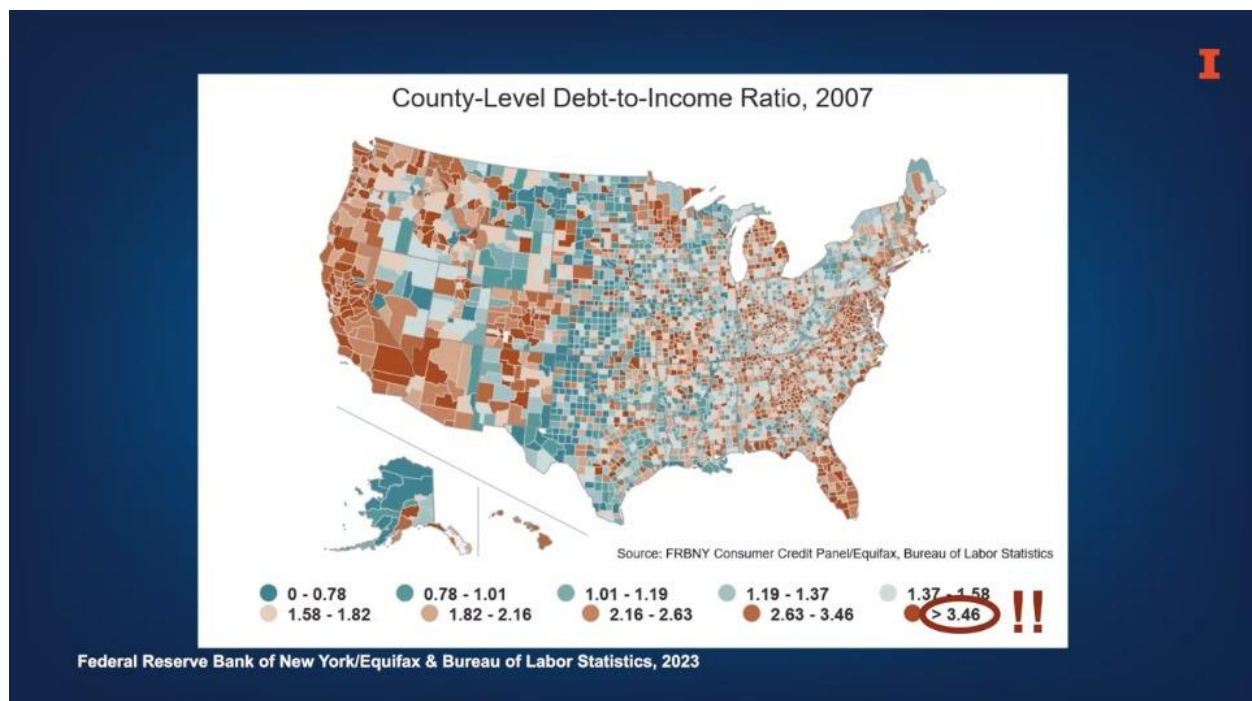
**Research Examining
FFR Increases**

- Large firms avoid disruption
 - Increase short-term borrowing
 - Maintain jobs, output, grow inventories
- Small firms cut inventory, jobs, output
- Economy shrinks

Does the data support these ideas? Well, yes and no. Research has examined how businesses respond to an increase in the federal funds rate and whether it matters for the economy as a whole. The research indicates that large firms are mostly unaffected when rates increase. They maintain their levels of production and employment when interest costs increase and revenue declines. These larger firms have no problems accessing commercial paper markets and other sources of short-term credit. This allows them to increase their short-term borrowing, even when profits decline. At the same time, the inventories of large firms grow. In sharp contrast, the transmission of monetary policy to the borrowing costs of small firms matters a lot more. Smaller, more financially constrained firms respond to an increase in interest rate costs by reducing inventory and by cutting work hours and production. Importantly for the economy as whole, the effect is still sizable. This is because smaller firms still employ more than one half of all workers in the United States. Moving on, let's now examine how households respond to monetary policy. That's not to say that all households are carefully reading statements from the FMC. However, their decision making will react to prevailing economic conditions, including changing market interest rates. The largest line item on a household's balance sheet is usually their mortgage.

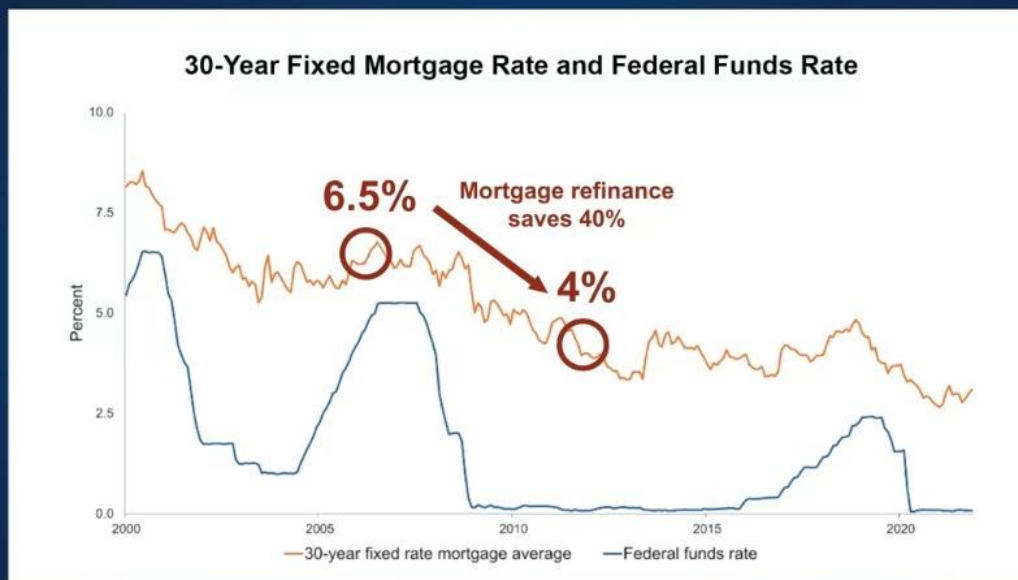


This graph shows you the interest rate on 30-year fixed rate mortgage and the federal funds rate. As you can see, the two rates move together so mortgage rates will react to changes in monetary policy.



How big of an effect can we expect from mortgage rate changes? The answer depends on how indebted households are. If households have zero debt, then changing borrowing costs will be irrelevant. The map shows you the median debt to income ratios

for households across all counties in the United States in 2007, right before the 2008 financial crisis. As you can see, many counties have households with debt that is more than three times their annual income. With such high borrowing relative to annual income, changes in the interest rate may have sizable effects on household finances. While some mortgages in the United States have an adjustable interest rate, most mortgages are fixed rate. When the FMC decides to lower the federal funds rate, mortgage rates tend to decrease as well. In response, households may decide to refinance their current mortgage. This means the household replaces the current mortgage with a lower interest rate mortgage. Also, this can involve upfront cost. The interest rate savings due to refinancing can be substantial. Let's go back to the craft with the mortgage interest rate but focus on the 21st century.



In 2006, the mortgage interest rate was about 6.5%, in December 2011, it had fallen below 4%. A household that had not paid any principle could save almost 40% on mortgage interest expenses by refinancing their mortgage, and a lot of people did. Recent research found that the average household that refinanced saved almost \$1,000 per month in mortgage payments.

Monetary Policy Transmission to Households

Decrease in FFR $\Rightarrow$ lower mortgage payments

Bigger effects when household:

- More indebted

- Refinances loan

Cash savings can stimulate consumption



What did the households do with the extra cash? Household that saved on mortgage payments tend to consume more. For instance, people may go out for dinner or start renovating their house. Spending on durable goods and cars increases. As a result, the demand for goods and services goes up, which benefits businesses and stimulates the economy.

Summary

Monetary policy impacts
businesses and households

Changes costs of borrowing

Lower rates can raise overall
demand for goods and services



Let's wrap up. In this lesson, we were specific about how monetary policy transmits to

the economy. When the Central Bank changes the interest rate, this can impact the interest expenses of businesses and households. We saw that the effects matter for small financially constrained businesses. We also saw how a lower policy rate reduces mortgage payments of households. In the aggregate, lower interest payments can stimulate greater consumption and investment. This means the overall demand for goods and services goes up.

Lesson 3-3: Limits of Monetary Policy

Lesson 3-3.1. Limits of Monetary Policy: Supply Shocks



Learning Objectives

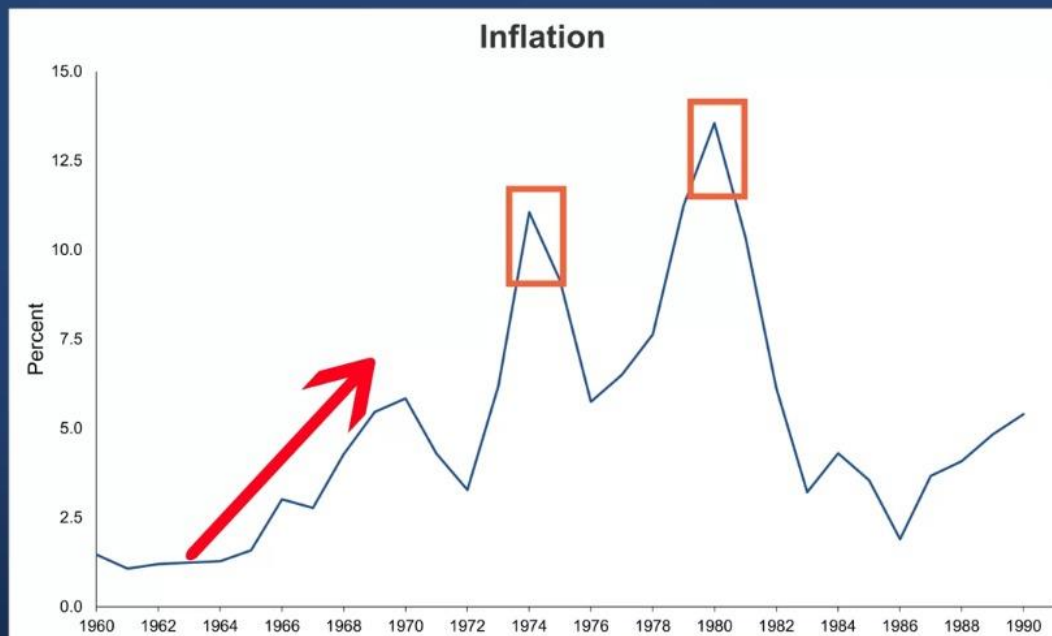
Case study: 1970s USA $\Rightarrow$ “stagflation”

- Low growth
- High inflation

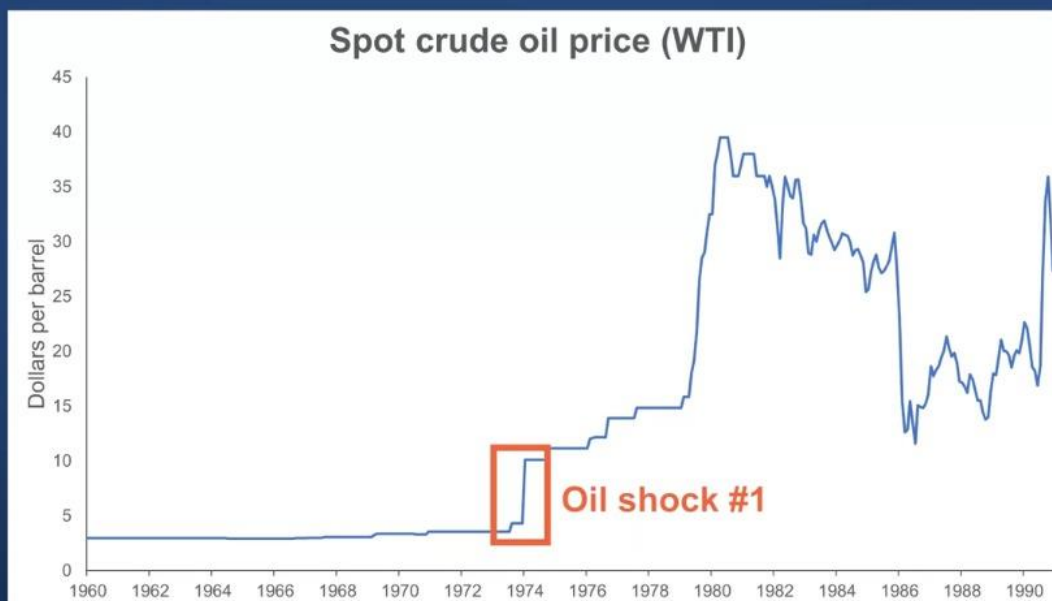
Limits of monetary policy

- Rate cuts stimulate growth ...
- but make inflation worse

Hello, and welcome to this lecture on stagflation. In this class, we consider the period in the 1970s that was marked by low growth, high inflation, and high unemployment. This case study will help us to better understand the tradeoffs and limits of monetary policy. The term stagflation is a combination of stagnation and inflation. Stagflation is particularly challenging for central banks. To stimulate the economy, central banks lower interest rates, however, low interest rates also tend to increase inflation. In a situation where growth is low, and inflation is high, central banks have limited options.



The graph shows you consumer price inflation from 1960 to 1990. You can see that during the late 1960s, inflation started to accelerate. The largest spikes are in 1974 and 1980, when the inflation rate hit 11 and 13.5 respectively. Let's look at the reasons for these spikes.



The main reason are two large oil shocks. The first oil shock was triggered by an oil embargo of OPEC against the United States and other mostly Western countries in the

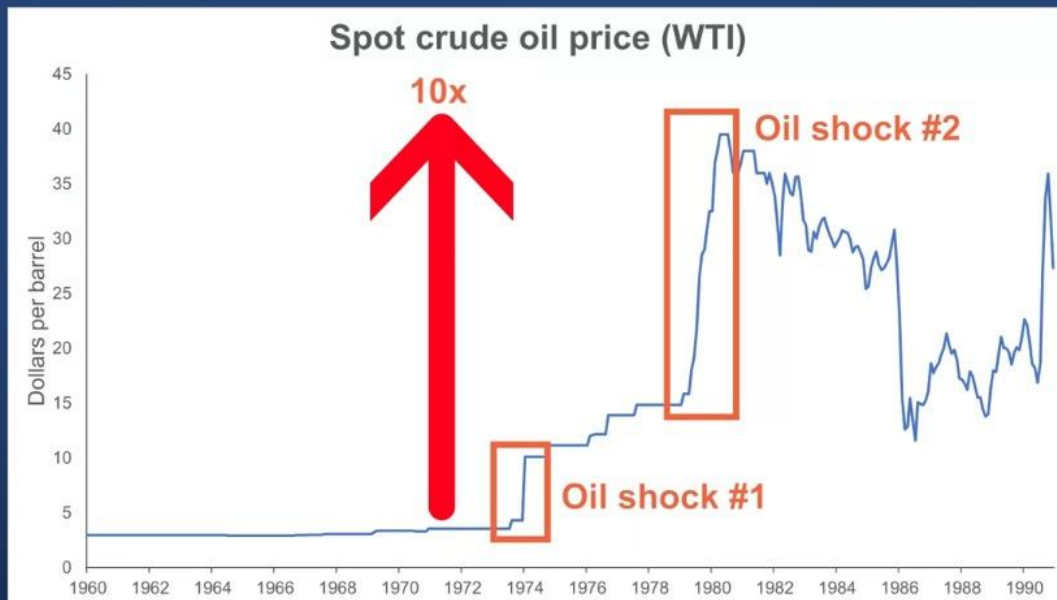
wake of the Yom Kippur War. Within a few months, the price of oil rose nearly 300 percent from three dollars per barrel to about 11 dollars.



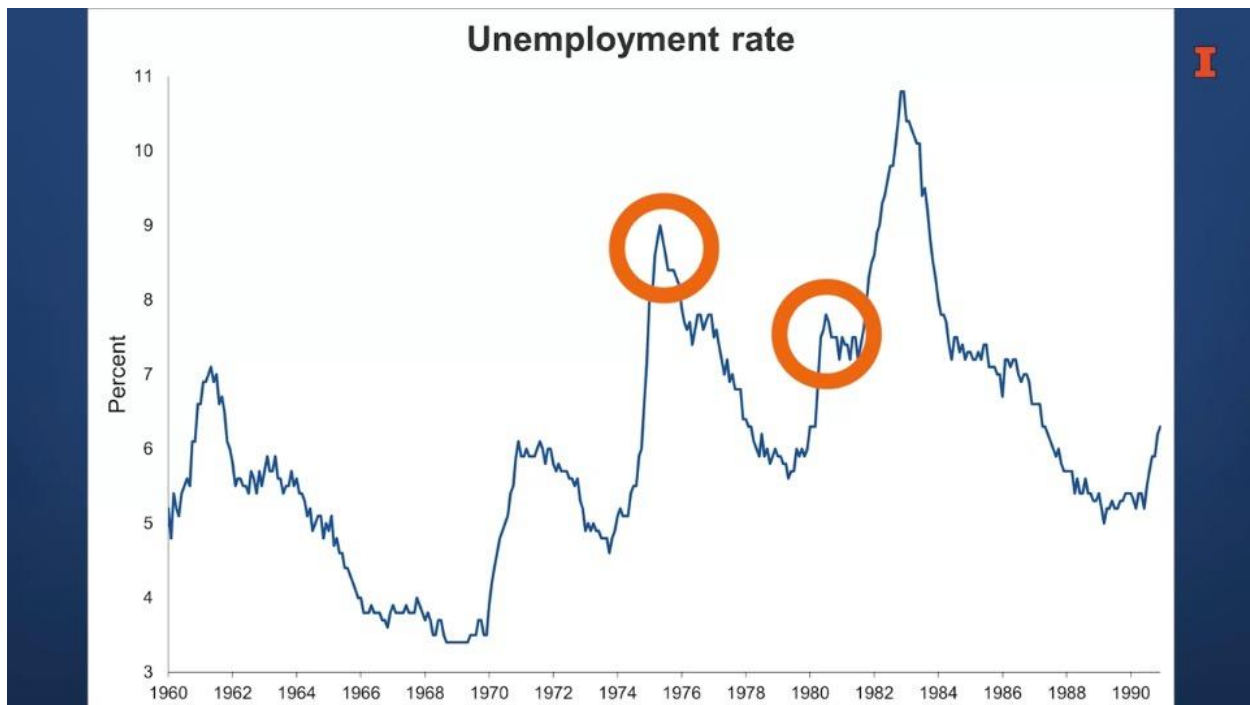
Impacts of Oil Shock

- Crude oil = production input
- Producer prices jumped
 - Gasoline, power, chemicals, etc.
 - Passed on to consumers
- Consumers reallocated spending

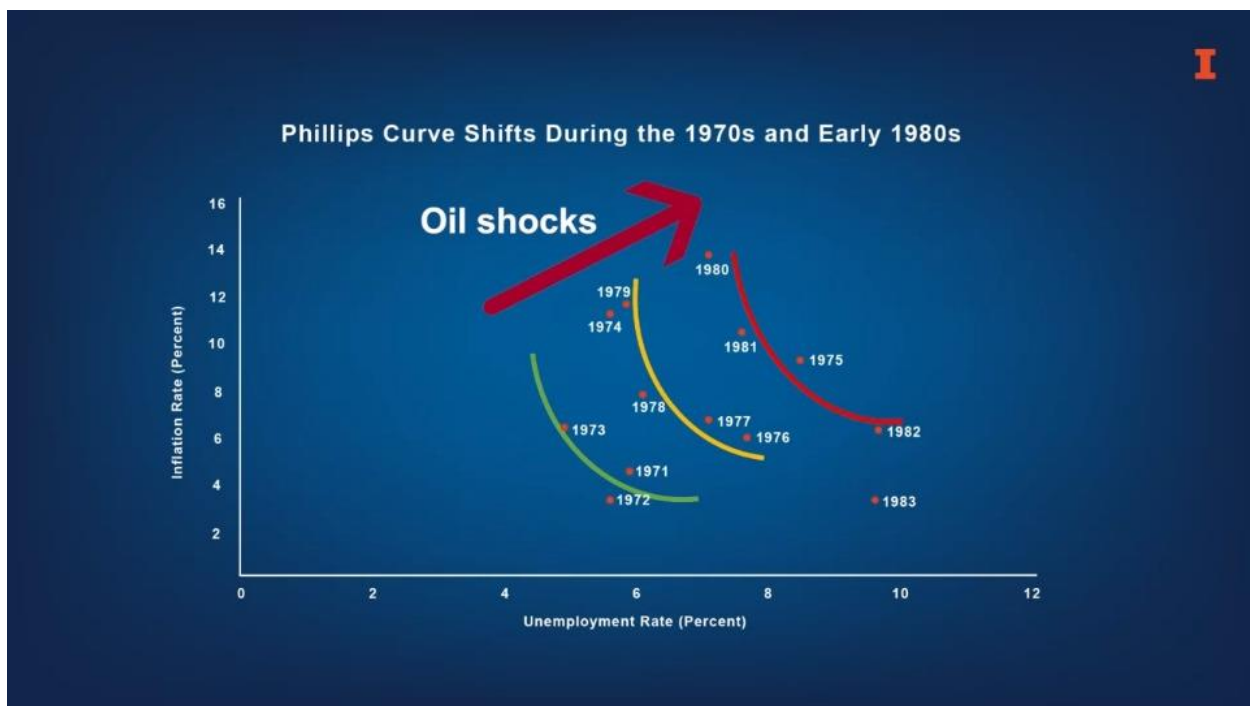
This was an enormous cost shock to the economy, since almost all parts of the economy used oil as an import, for instance, in form of gasoline or for power generation or in the chemical industry, producer prices increased drastically. At the same time, consumers had to pay more for gas and heating, reducing their spending on other consumption goods.



Just a couple of years later, the second oil shock hit the economy. In the wake of the Iranian revolution in 1979 and the beginning of the Iran-Iraq War in 1980, oil production declined. As a result of declining oil supply and uncertainty about future oil supply, oil prices skyrocketed. In 1980, the price of oil reached 39 dollars and 50 cents per barrel. A more than tenfold increase since early 1973. In the US, the car manufacturers were particularly hard hit as they struggled to offer more fuel-efficient cars. These large shocks led to a slowdown of the economy.

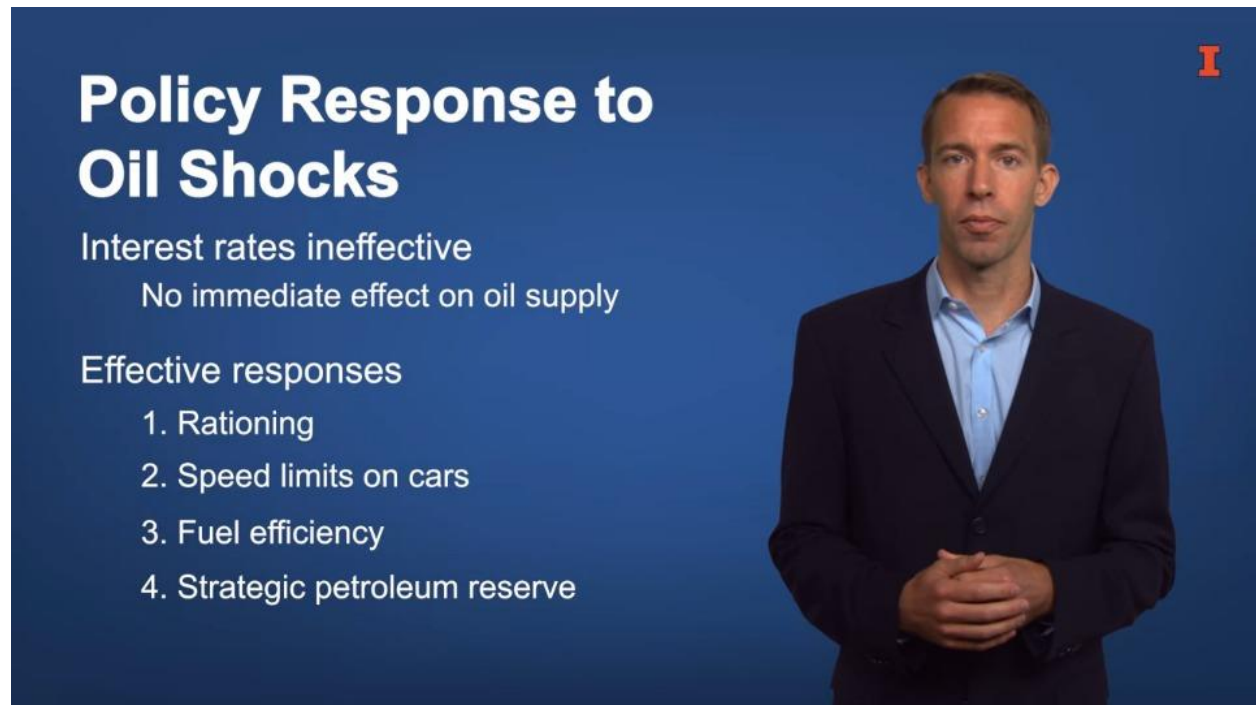


In the graph you see the unemployment rate. It reached a post-World War II high of nine percent in 1975 and decreased again with the second oil shock in 1981, reaching 7.8 percent. Recall that the Federal Reserve's mandate is stable prices and maximum employment. But the oil shocks produce the opposite effects, high inflation, and high unemployment. The oil shocks are examples of supply shocks.



The craft illustrates how supply shocks shift out the tradeoff between unemployment

and inflation, which is also called the Phillips Curve. We discuss the Phillips Curve in detail in a separate lecture.



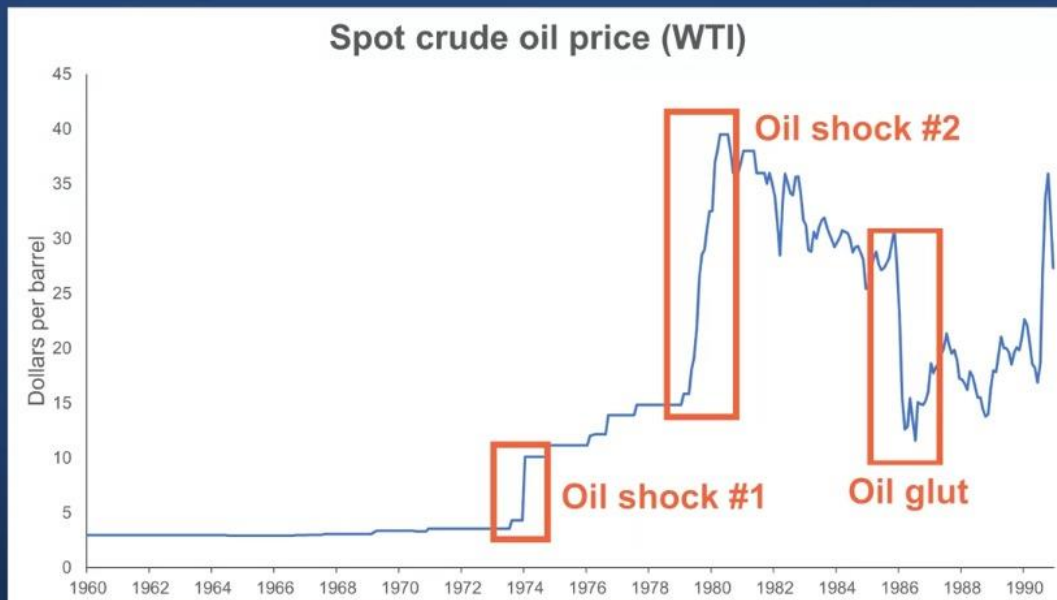
Policy Response to Oil Shocks

Interest rates ineffective
No immediate effect on oil supply

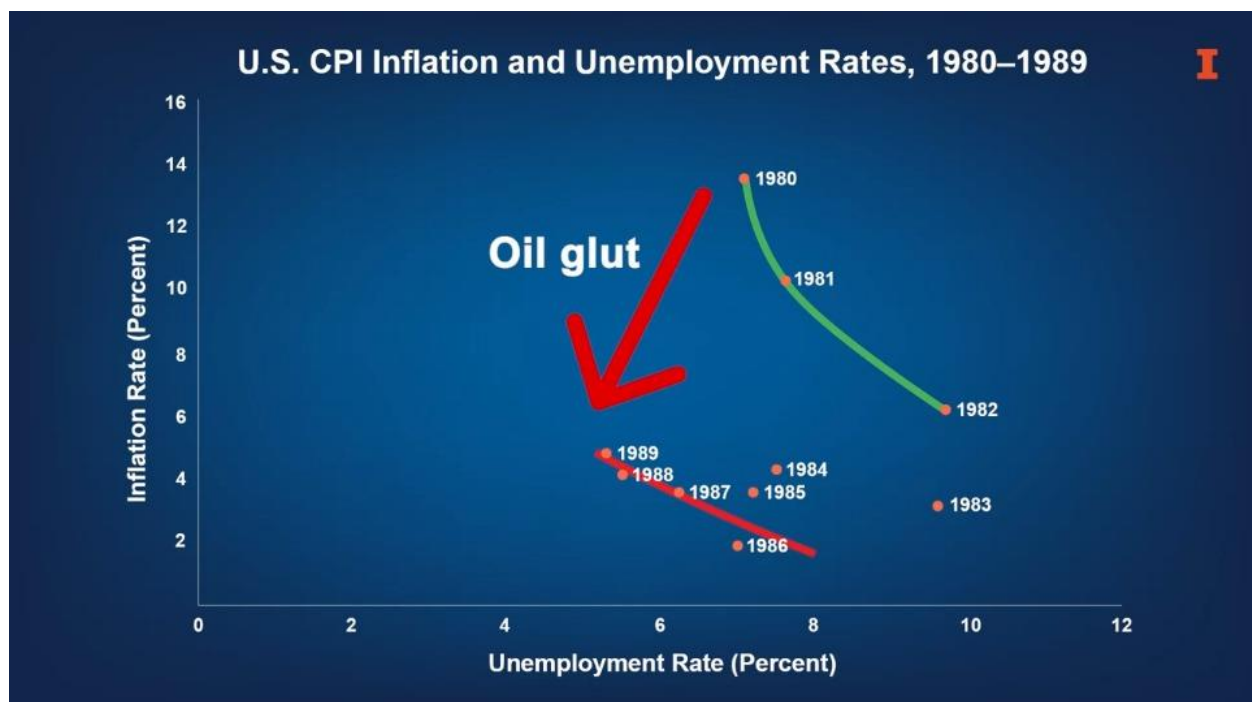
Effective responses

1. Rationing
2. Speed limits on cars
3. Fuel efficiency
4. Strategic petroleum reserve

When facing supply shocks, the options of monetary policy are limited as lower interest rates do not affect supply constraints. Indeed, many responses to the oil shocks were not monetary, but affected oil demand in the short and in the long run. The most drastic short-term response to the 1973 oil shock was rationing gas in the United States. For instance, vehicle with license blades having an odd number could buy gas only on odd numbered days, while others could buy only on even numbered days. Responses with considerable long-term impact on oil demand eventually helped to reduce oil prices. Here are a couple of examples. To reduce gas consumption, a national maximum speed limit of 55 miles per hour was imposed in 1974. A year later in 1975, the corporate average fuel economy standards were created. These standards required better fuel economy for cars and light trucks in the US. At the same time, the US developed the Strategic Petroleum Reserve.



These measures and increases in production eventually led to the so-called oil glut in the 1980s. Oil prices dropped considerably and contributed to bringing down inflation.



These supply shocks in the 1980s help shifting back the tradeoff between inflation and unemployment. As you can see in the graph, in the second half of the 1980s, considerably less inflation was needed to reduce unemployment when compared to the early 1980s.

Oil Shocks and Inflation Expectations

Worker beliefs about inflation

Before 1970: low

In 1970s: will remain high

Risk: greater wage demands

Cause further wage demands

“Wage-price spiral”



Even so, monetary policy can do little to counter its supply shocks. This period caused a change in inflation expectations. That is, the consensus shifted from a low inflation regime to a regime in which inflation would remain high.

Oil Shocks and Inflation Expectations

Worker beliefs about inflation

Before 1970: low

In 1970s: will remain high

Risk: greater wage demands

Cause further wage demands

“Wage-price spiral”

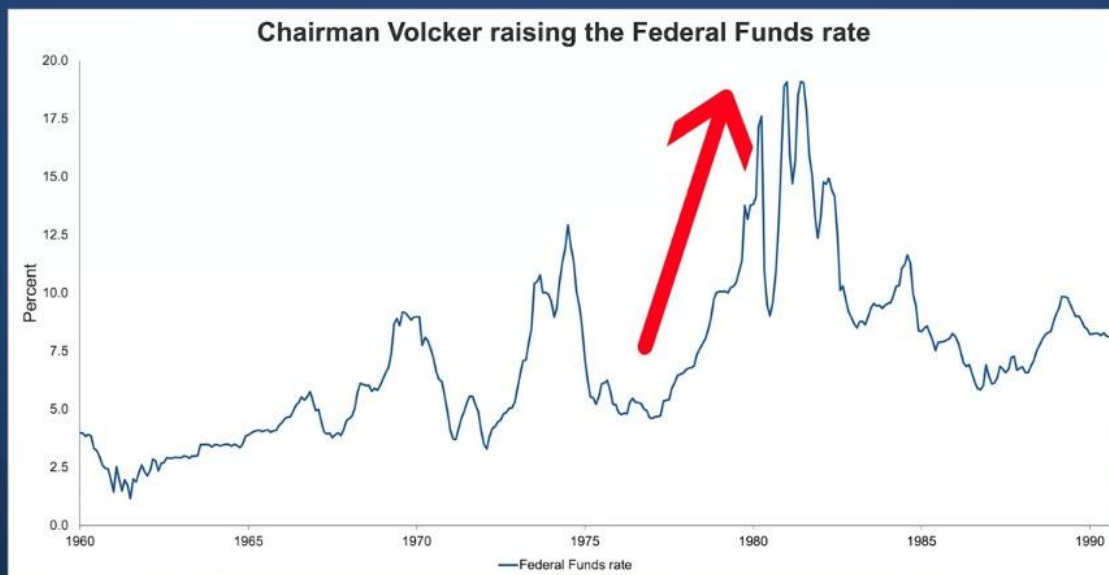


High inflation expectation led to high wage demands, which in turn increase inflation.

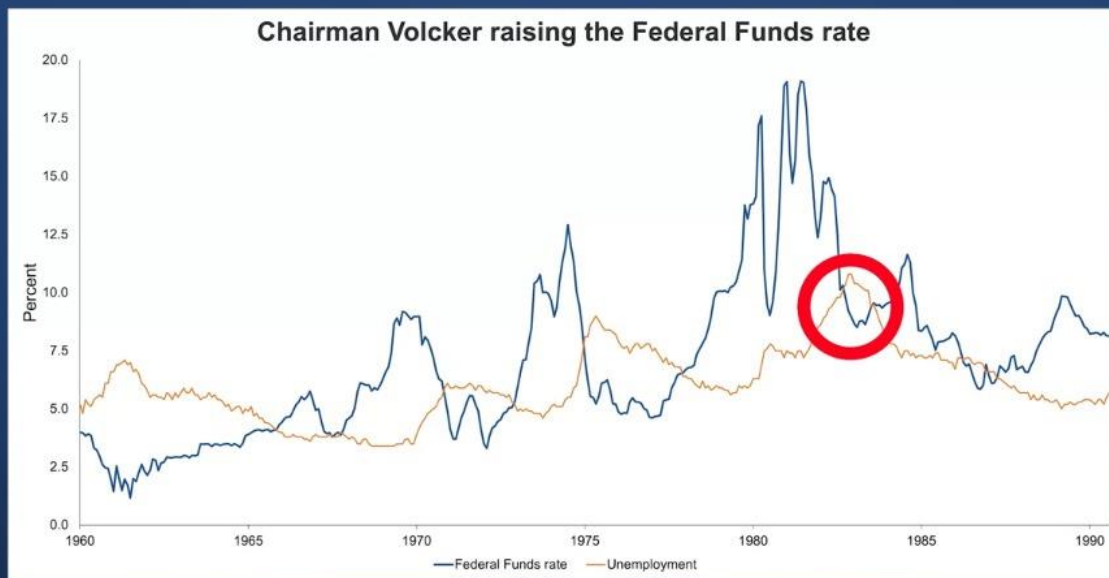
Paul Volcker



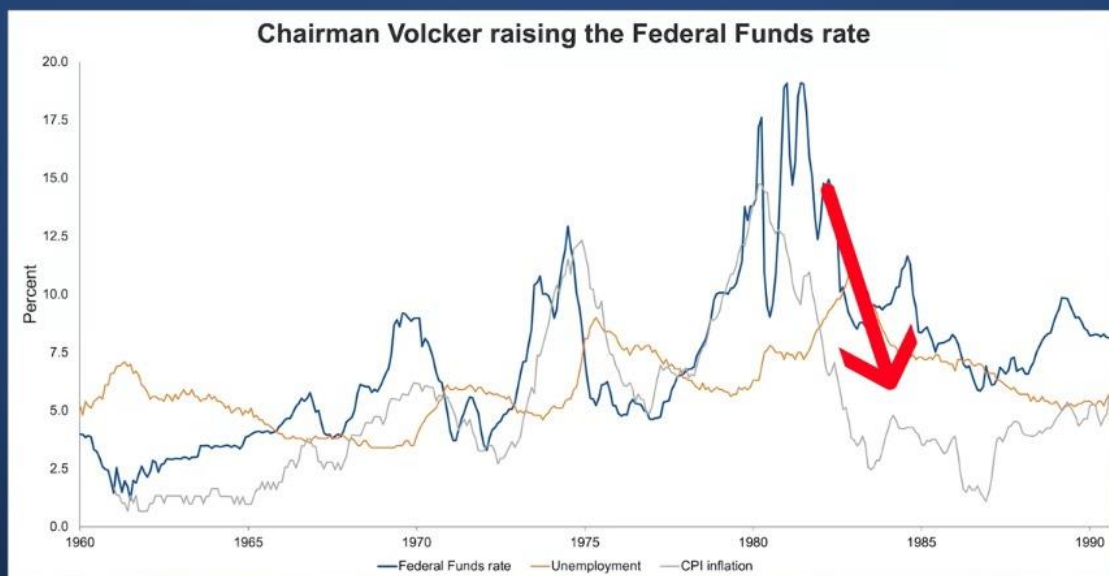
When Paul Volcker became the Chairman of the Federal Reserve in 1979, he wanted to bring down inflation and inflation expectations.



He did so by raising the Federal funds rate, which reached 19 percent in December 1981. The sharp increases in interest rates, together with increasing oil prices, triggered the 1980 and 1982 recessions.



Unemployment rose to over 10 percent the highest level between the end of World War II and the COVID-19 pandemic. In other words, in this period, the Federal Reserve chose to focus on inflation rather than unemployment despite the dual mandate of stable prices and maximum employment, raising the federal funds rate led to a large drop in inflation.



More importantly, the Federal Reserve accepted the price of temporary higher

unemployment to bring down inflation. This was seen as a credible commitment to low inflation in the future. This commitment in turn, brought down inflation expectations, which helped to establish a low inflation regime.

Summary

Supply shocks shift inflation-unemployment trade-off

Monetary policy does not impact supply shortages

Federal Reserve can manage long-term inflation expectations ...

... but this can be costly (high unemployment)



What have we learned in this lesson? First, supply shocks, such as oil price shocks, can shift the tradeoff between inflation and unemployment. Second, monetary policy is not an effective tool to counteract supply shocks. Third, if supply shocks move inflation expectations permanently, central banks may have to accept high unemployment temporarily, to bring down inflation and inflation expectations.

Lesson 3-3: Limits of Monetary Policy

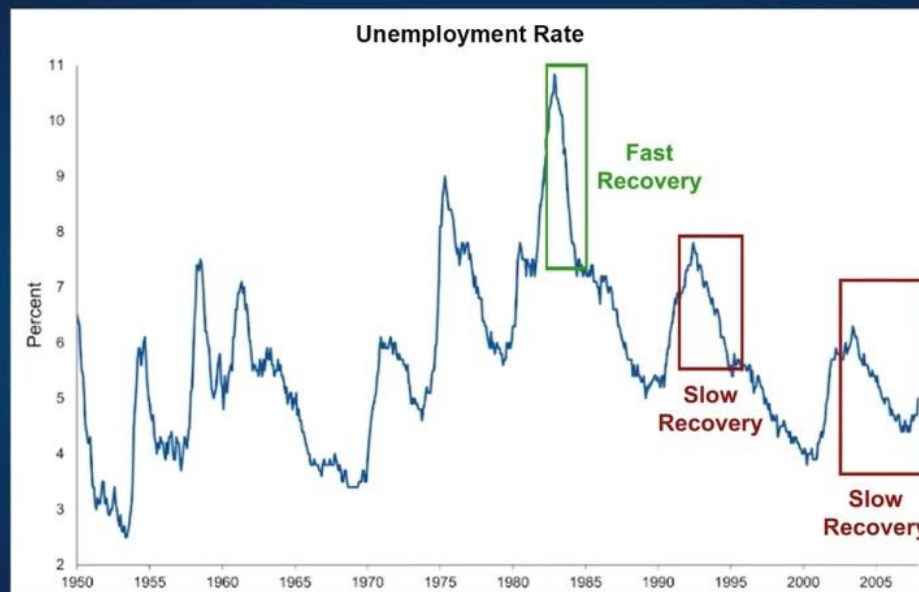
[Lesson 3-3.2. Limits of Monetary Policy: Jobless Recoveries and Risk- Taking Incentives](#)



Learning Objectives

- Limits of monetary policy
- “Jobless recovery”
- Low interest rates ⇨ financial fragility

Hello, and welcome to this lecture on the limits of monetary policy. In this class, we will consider times when monetary policy hasn't worked very well. We will examine two related issues. First, the idea of a jobless recovery. This term was first used to describe the recovery from the 1991 recession. Despite significant cuts to interest rates by the Federal Reserve, the labor market recovered more slowly than expected. We will try to understand why this can happen. We will then consider an unintended consequence of loose monetary policy excessive risk-taking. Lowering interest rates can stimulate the economy. However, what if rates are kept too low for too long? Some argue that this might cause financial instability. We will examine this argument in detail. Let's get started.



Historically, unemployment had dropped sharply after recessions ended. Look, for instance, at what happened after the recession of 1982. In contrast, after the 1991 recession, the unemployment rate came down only gradually. This was despite the Federal Reserve keeping rates low for three years and GDP growing by over 5% between 1992 and 1995. This slow recovery in the labor market puzzled economists. The same pattern emerged after the recession in 2001. The Federal Reserve cut the federal funds rate from 6.5% to 1% in response to the recession, and it did not increase the federal funds rate until the summer of 2004. Unemployment, however, did not start to fall until the summer of 2003, and even then, the decline was only gradual. This type of monetary policy accommodation at such low interest rates was unprecedented. For this reason, economists now refer to the early 2000s as the jobless recovery.

2001 Jobless Recovery

Problem: slow employment growth
despite large FFR cuts

What happened?

Jobs data:

1. High youth labor participation
2. Longer unemployment duration
3. Fewer job vacancies

⇒ Firms not hiring



Let us examine first what the data tell us about the labor market and the 2001 jobless recovery. Indicators for the labor market suggested that workers faced significant challenges finding jobs. First, labor force participation, particularly among younger workers, declined. Second, unemployment duration increased significantly. Third, a popular index of help wanted advertising, a measure of job vacancies, remained below its recession levels. The evidence suggests that firms were just not hiring as much as expected.

2001 jobless recovery

Problem: slow employment growth
despite large FFR cuts

What happened?

Survey evidence:

By 2003, 2.8m jobs lost

2.4m manufacturing jobs

⇒ Mostly manufacturing jobs



In addition, the current employment statistics survey showed that as of the end of September 2003. Non-farm payroll employment had fallen by some 2.8 million jobs since the beginning of the recession in March 2001. The manufacturing sector was by far the hardest hit sector. About 2.4 million of the 2.8 million jobs lost were in the manufacturing sector. Let's dig deeper.

2001 Jobless Recovery

Problem: slow employment growth
despite large FFR cuts

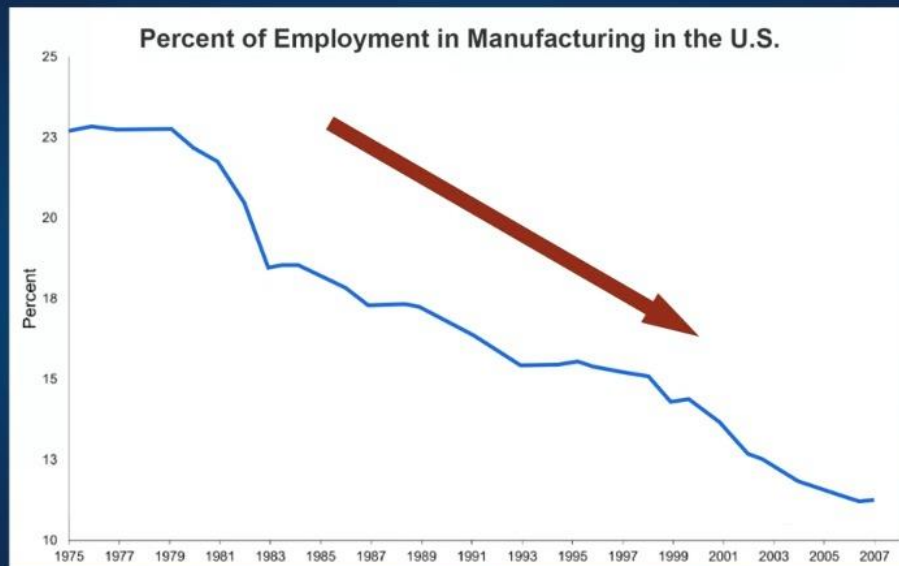
Leading explanations:

1. High uncertainty about future
2. Structural change



Why was the labor market response so slow? And why was the manufacturing sector hit

so hard? There are two main explanations. The first is high uncertainty about the economy. This makes firms reluctant to invest or hire. In the early two thousands, the most obvious source of uncertainty were the September 11 terrorist attacks and the aftermath, including the wars in Afghanistan and Iraq. Second, the manufacturing employment losses were due to structural changes in the US economy. Structural change means the permanent decline of certain industries, the decline of manufacturing was not new.



The share of workers employed in the manufacturing sector collapsed beginning in the early 1980s. This was partly driven by international competition.

2001 Jobless Recovery

Problem: slow employment growth
despite large FFR cuts

Leading explanations:

1. High uncertainty about future
2. Structural change



Most notable was the increased competition in the auto market, where Japanese and Western European automakers expanded into the US market. This happened after the 1980 oil crisis. Structural change is usually a slow process. This makes the consequences hard to gauge for policymakers in real time. When you look back at the discussions of the early 2000, one factor is notably absent. On December 11, 2001, China became a member of the World Trade Organization. As a result, US trade policy eliminated potential tariff increases on Chinese air imports. Recent research has found that shop losses in manufacturing were concentrated in US firms that faced import competition from China.

Limits of Monetary Policy

Can monetary policy help with ...

Uncertainty?

Possibly

Eliminate monetary policy
uncertainty with forward guidance

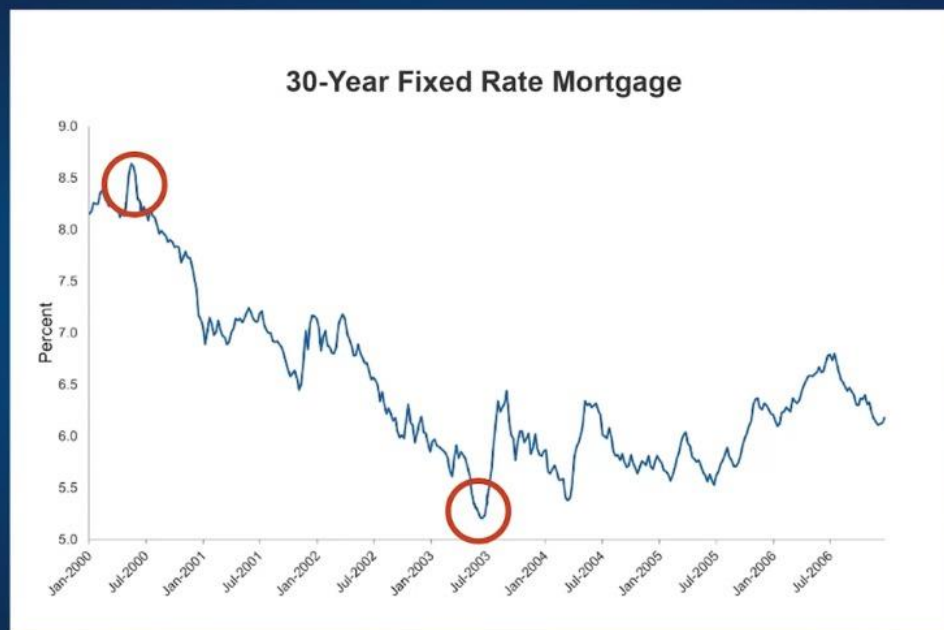
Structural change?

Probably not

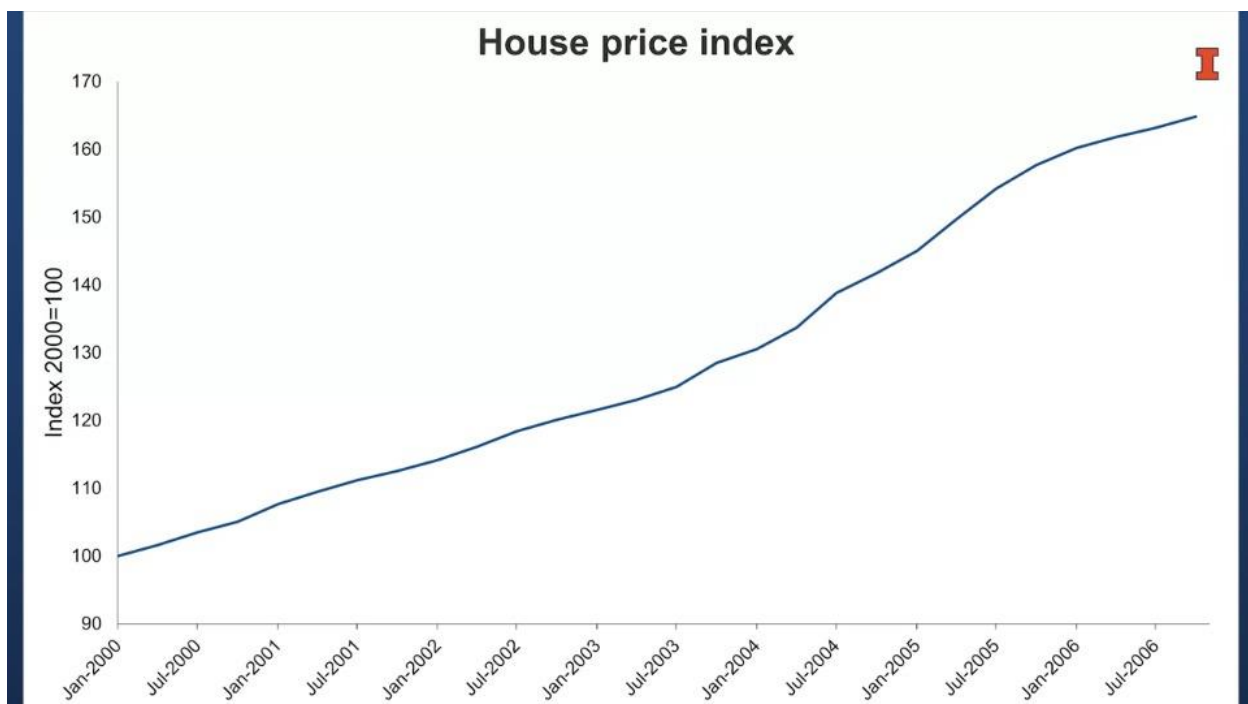
Monetary policy addresses
cyclical unemployment



Can monetary policy be effective when uncertainty is high or structural change is underway? In general, monetary policy can reduce uncertainty about future economic developments by promising accommodative monetary policy now and in the future. This involves the use of forward guidance. However, in the case of the early 2000s, the uncertainty was caused by terrorist attacks and subsequent wars. Monetary policy probably could not reduce this type of uncertainty. Similarly, monetary policy can do little to reduce the employment effects of structural change. We would expect jobs lost in declining industries to be replaced eventually with jobs in growing industries, for instance, in the service or tech industry. However, this process can take time. New jobs need to be created and workers need to relocate and retrain. Monetary policy works best when addressing cyclical fluctuation. It can do little when structural change drives unemployment. In the remainder of this session, let's consider the risks of accommodative monetary policy. In 2001, Paul McCully, an economist at the investment management firm PIMCO, predicted that the extremely low federal funds rate set by the Federal Reserve could cause house prices to increase.

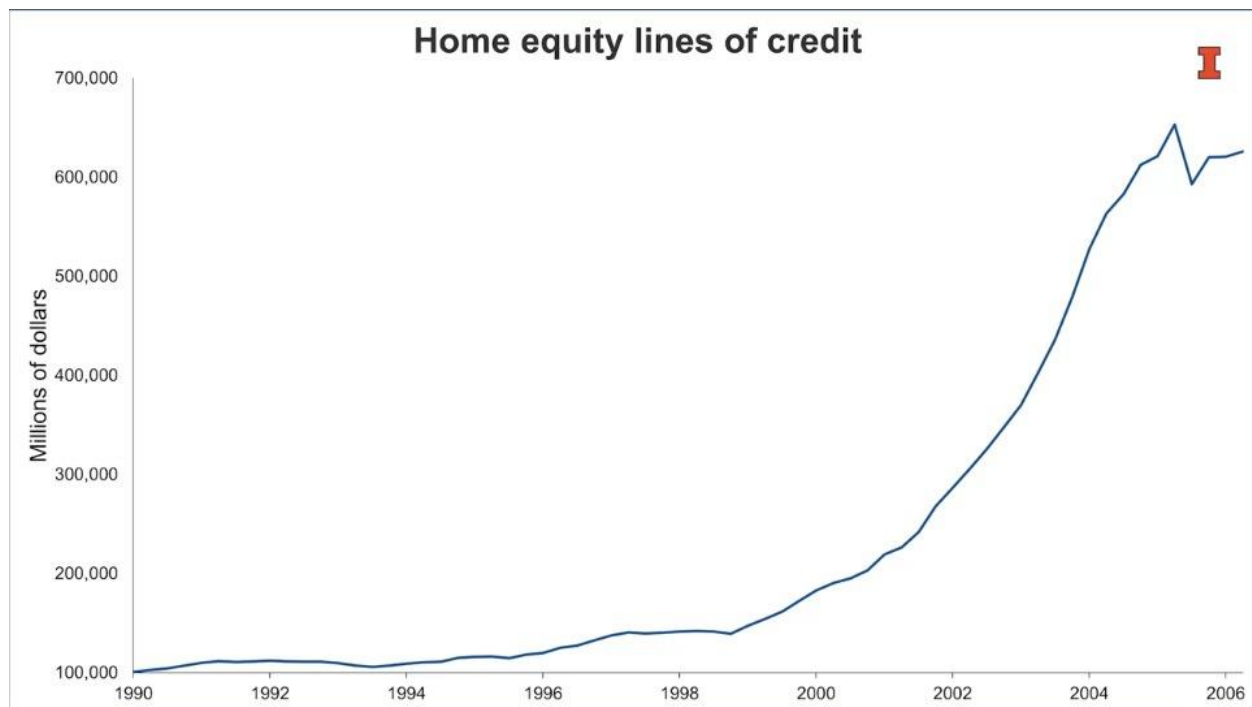


As the graph shows, the mortgage interest rate dropped from over 8% in 2000 to 5% in the summer of 2003.



Lower mortgage interest rates mean that people can afford new or larger houses. This increases demand for houses and house prices. This graph shows that house prices increased by over 60% between 2000 and 2006. As a result of rising house prices construction boomed. Due to a wealth effect, homeowners used mortgage refinancing

and home equity lines of credit to borrow more against the house. The proceeds were then spent on consumer goods, for instance, cars and appliances.



As you can see in the graph, home equity lines of credit increased from about 150 billion in 2000 to 650 billion by late 2005. This is a dramatic increase in credit supply. It viewed home construction, consumer spending and job creation.



Raghuram Rajan



By 2005, then IMF chief economist Raghuram Rajan expressed concern about a housing bubble. He argued that low interest rates and financial innovation created a situation in which financial market participants took more and more risk. He concluded that this risk could threaten the stability of the financial system. In hindsight, we know that his analysis was spot on.

Limits of Monetary Policy

Low interest rates $\Rightarrow$ excess risk

1. Lower borrowing costs
2. Higher leverage
3. Asset price bubbles

Central banks monitor financial stability



The more general point to keep in mind is that low interest rates can lead to a buildup of

risk. Lower interest rates can encourage excessive borrowing. This can fuel an asset price bubble in the housing market and financial markets. The link between low interest rates and risk-taking has been shown not only for the US, but for all major economies. Encouraging risk-taking is one way to stimulate the economy. However, central banks need to be mindful of the effects of accommodative monetary policy on the overall soundness of the financial system. Since the 2008 financial crisis, financial stability has been added to the Federal Reserve's mandate. The Federal Reserve now monitors the buildups of risks in the economy.

Summary

Monetary policy might not increase employment:

1. Permanently declining industries
2. External shocks (e.g., disasters)

Unintended consequences of low rates:

1. Inflated asset prices
2. Excessive risk-taking



Let's wrap up. We discussed two potential limits of monetary policy. First, monetary policy can do little when structural change or uncertainty about external events drives up unemployment. Second, loose monetary policy following the jobless recovery in the early 2000s may have fueled a house price boom in the United States. This experience highlights a more general problem. Lower interest rates can encourage risk-taking and can lead to a buildup of risk in the economy.